Catch Me if You Can: An Optimal Pest Survey Strategy to Delimit Biological Invasions

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CORS 2026

June 9, 2026

Delimiting Surveys



Figure: The spotted lanterfly and reported sightings (2026)

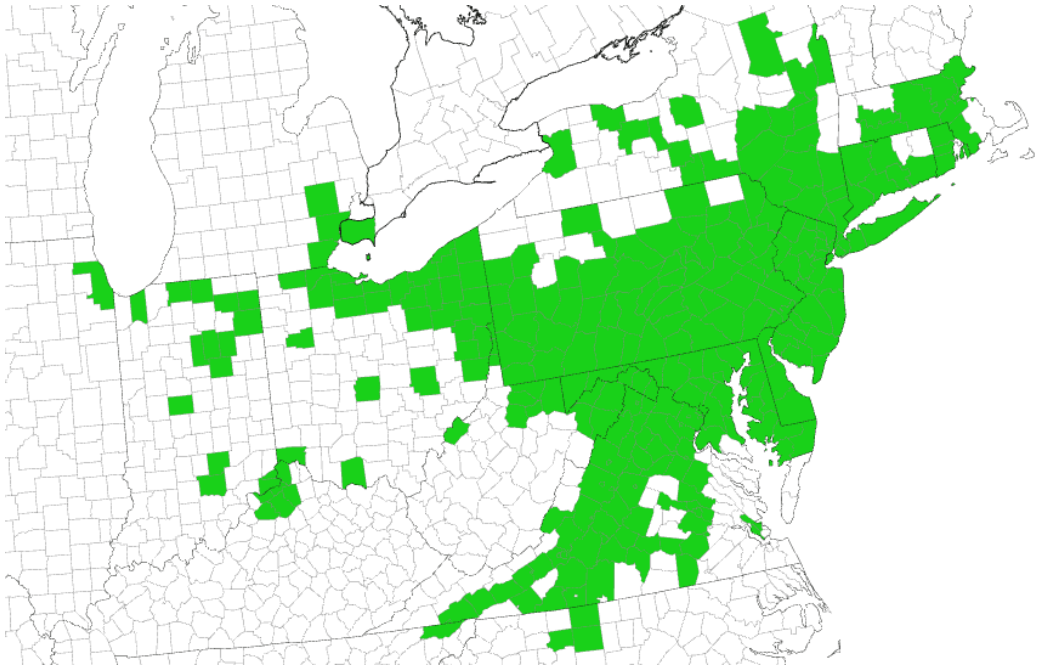


Image credit: Early Detection & Distribution Mapping System, Spotted Lanternfly

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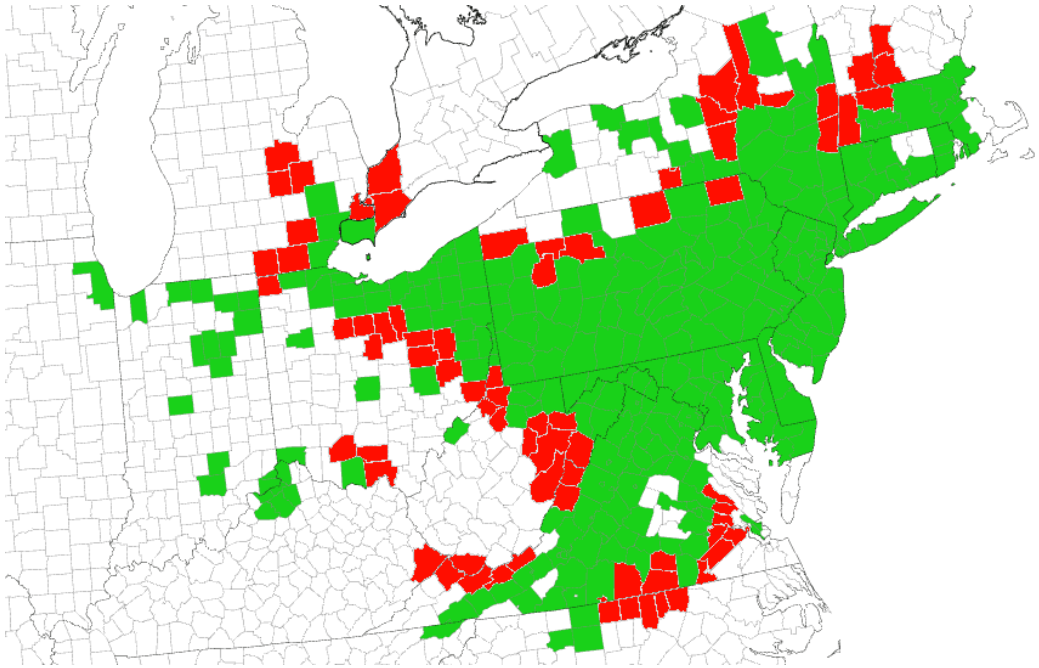
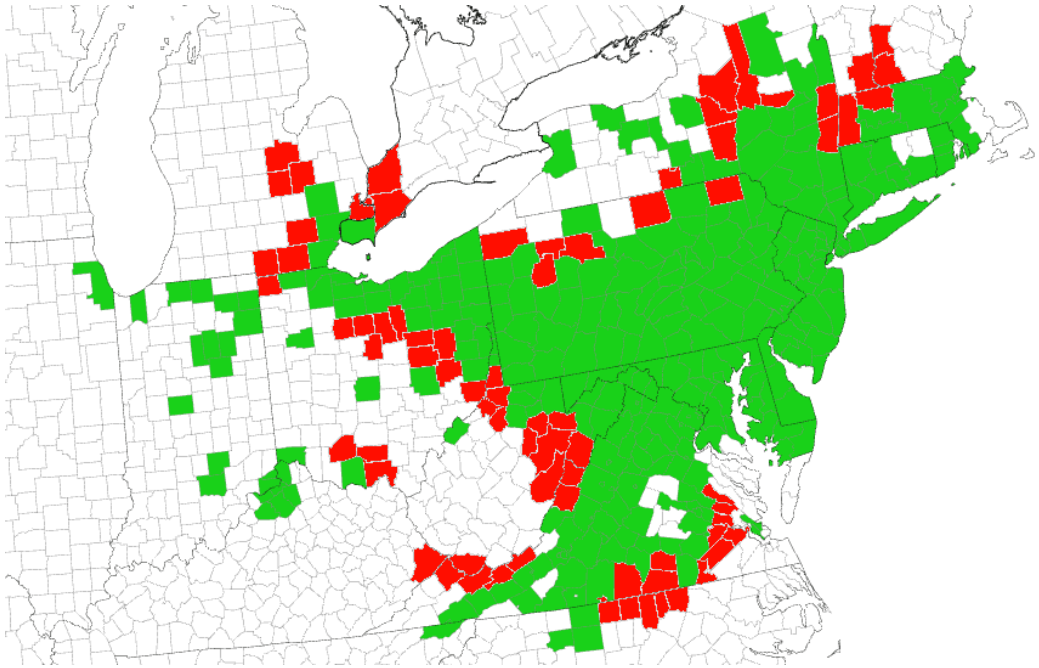


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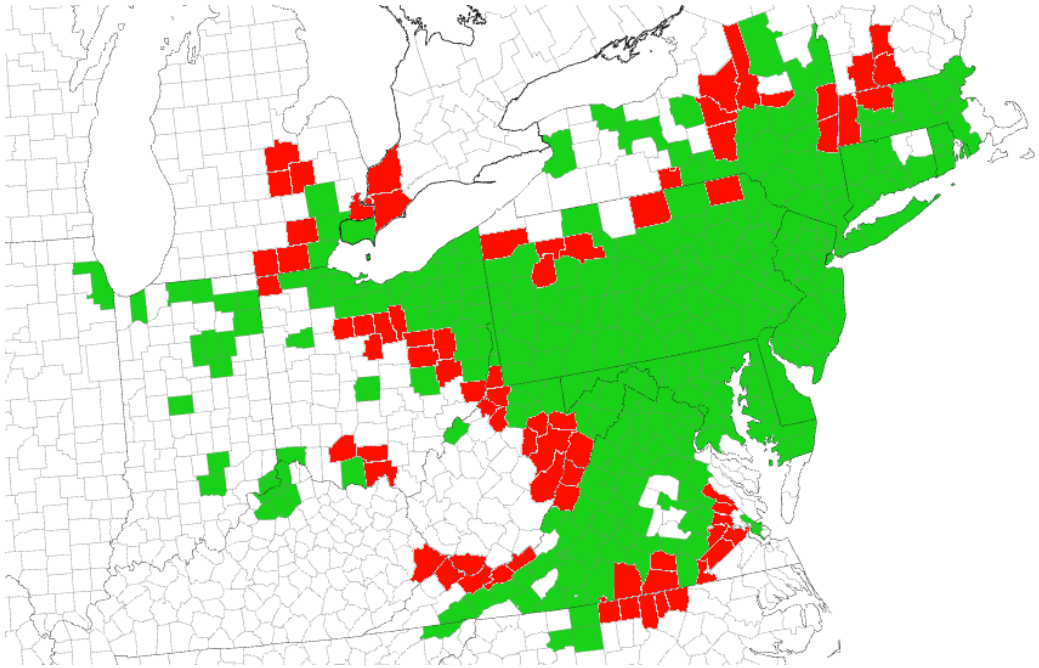
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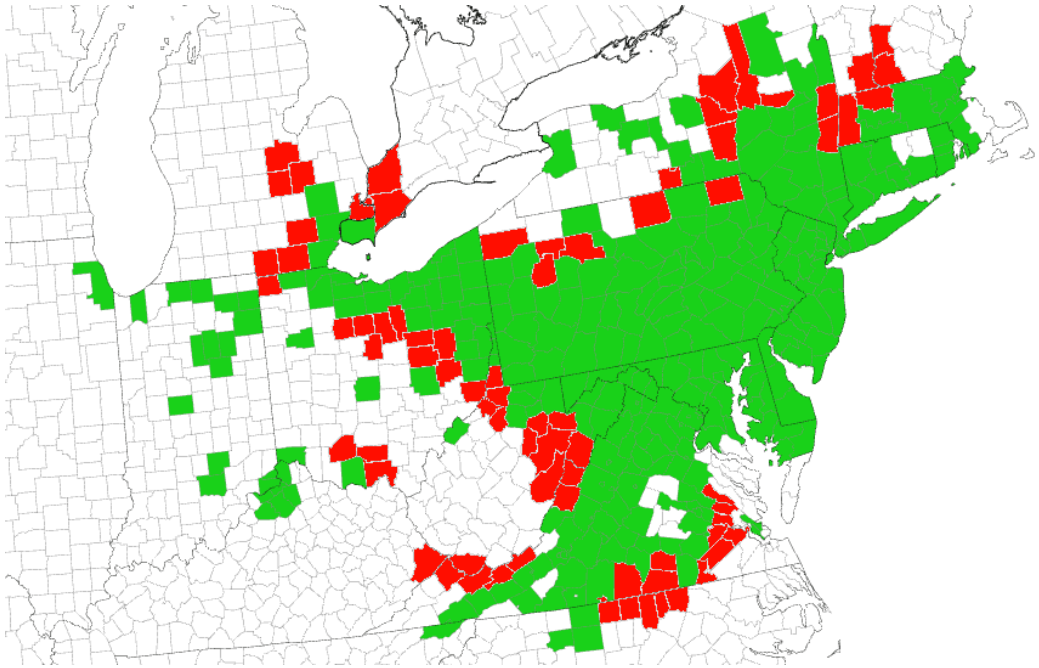
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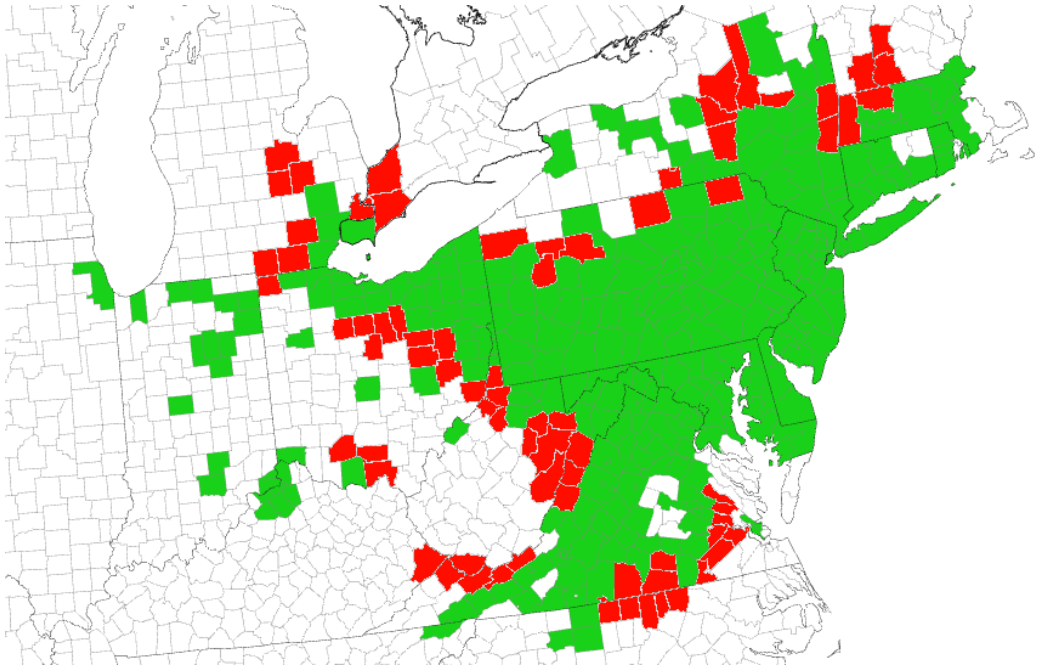
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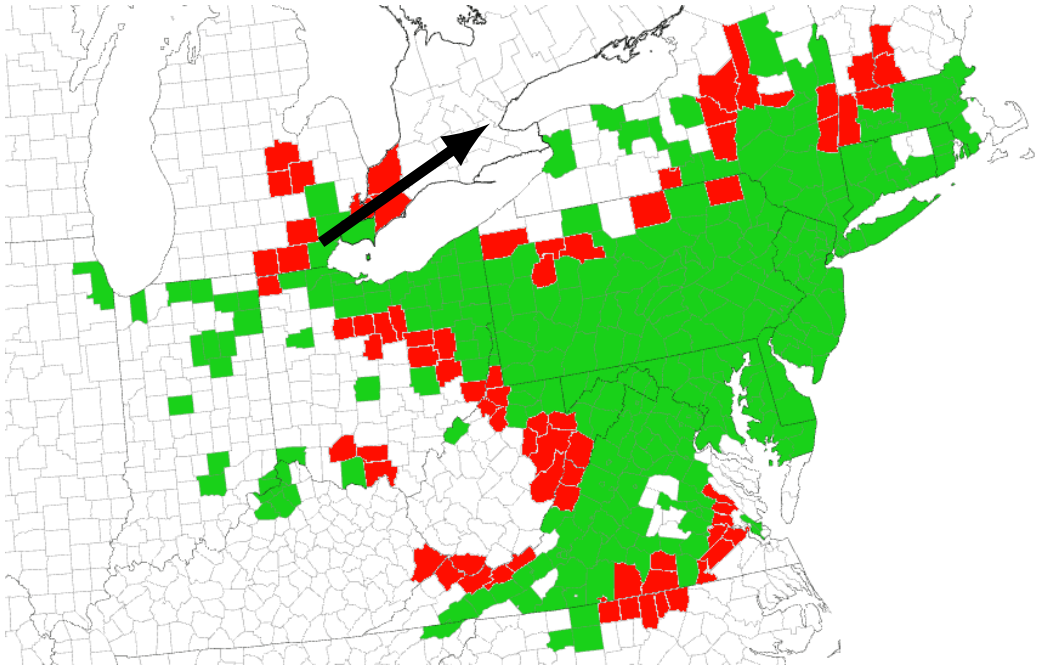
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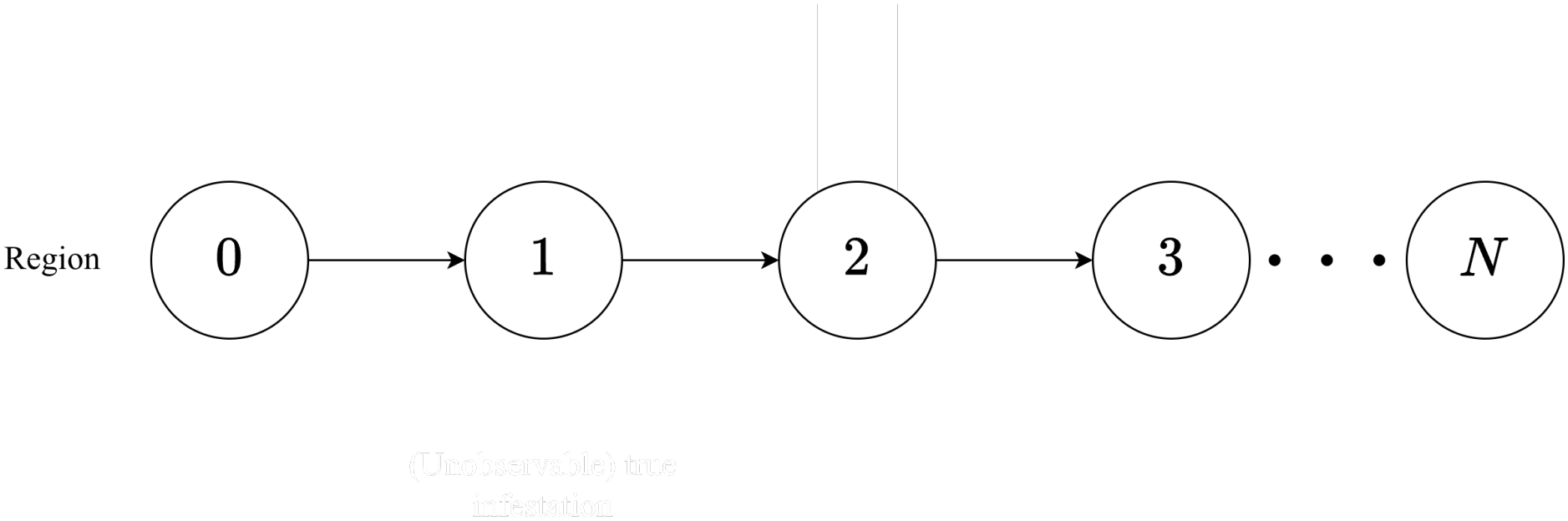


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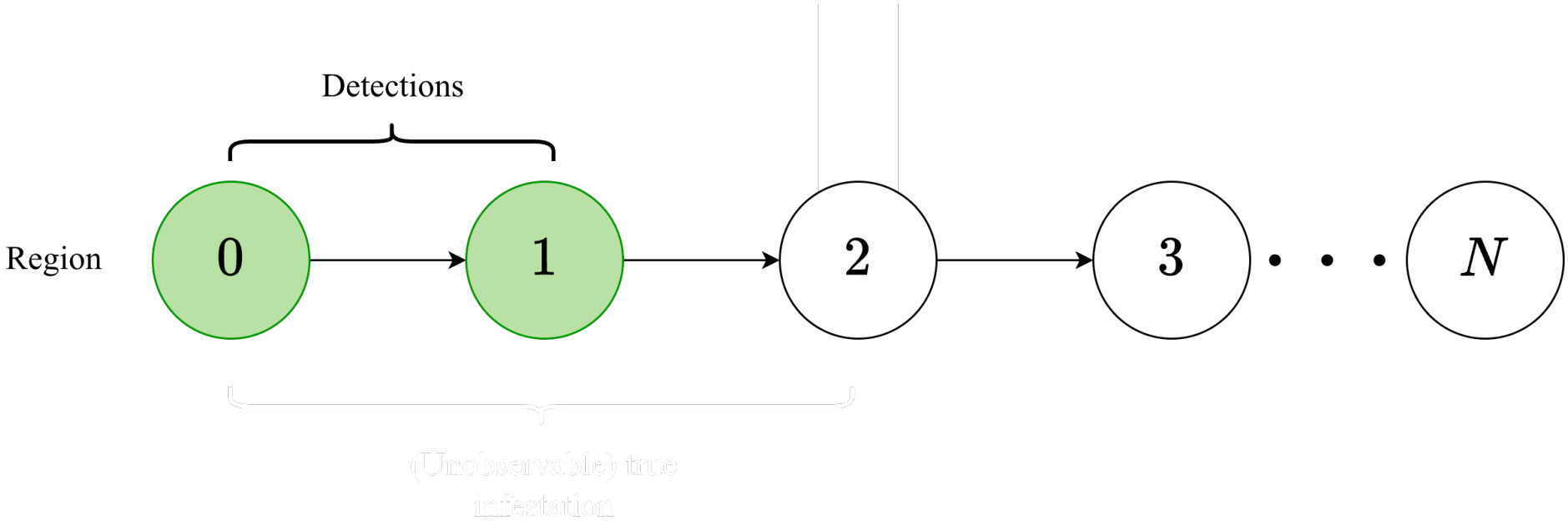
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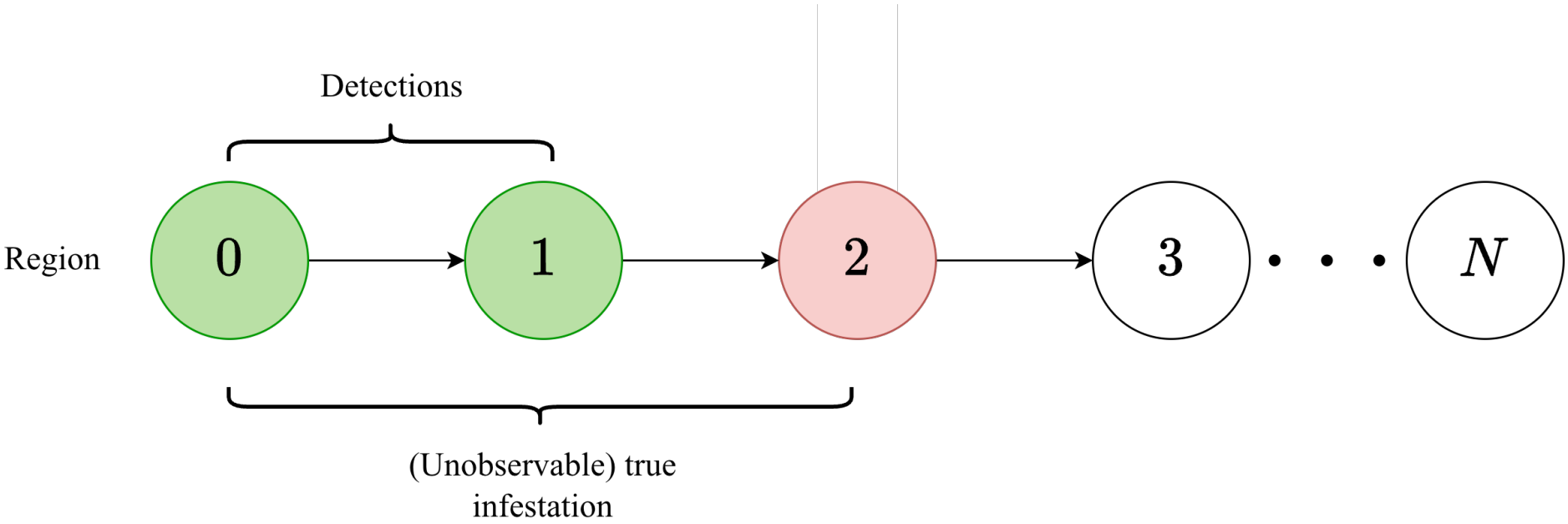
Delimiting Surveys - 1D Problem Formulation



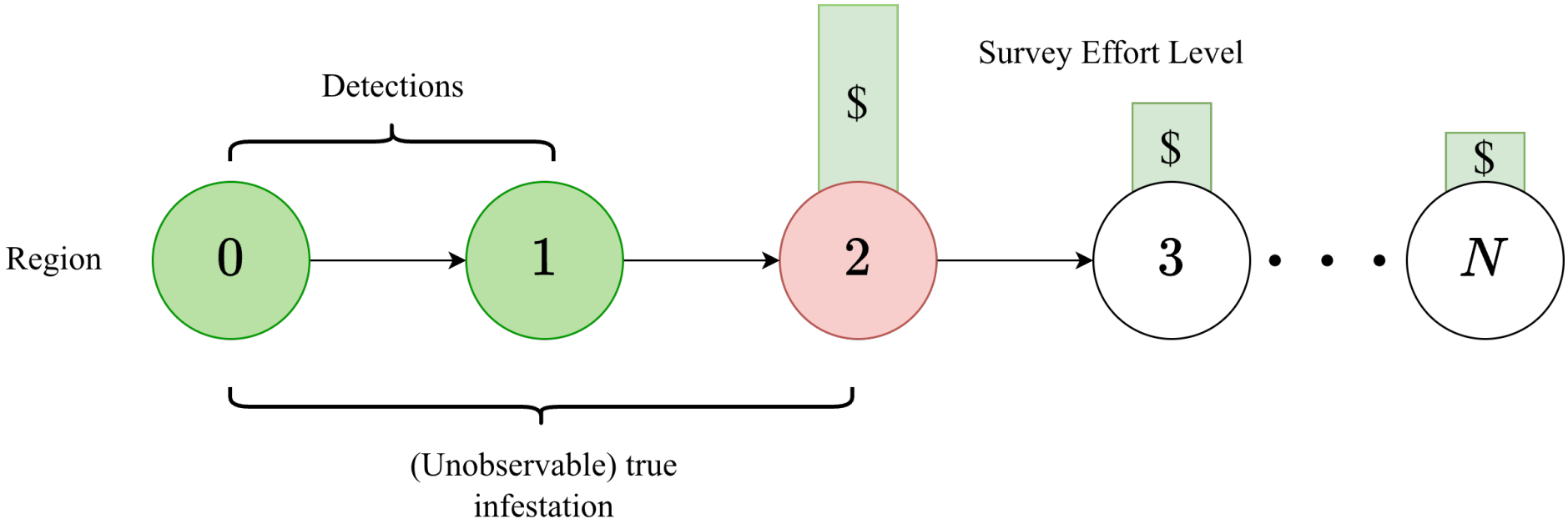
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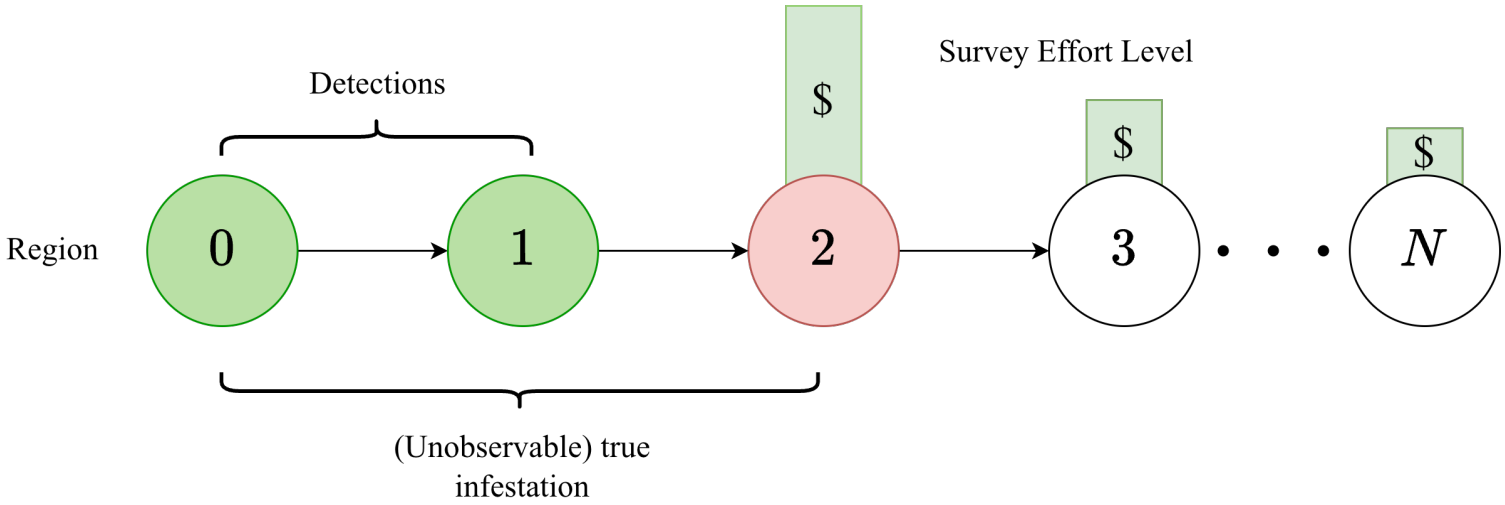
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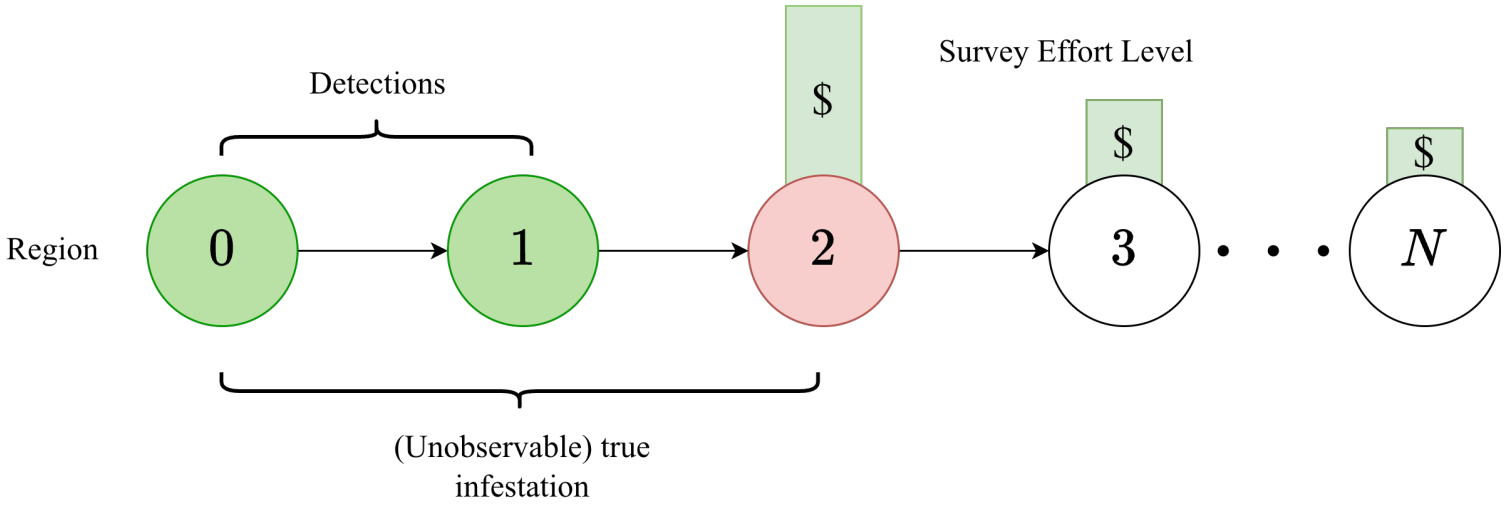


Problem Formulation



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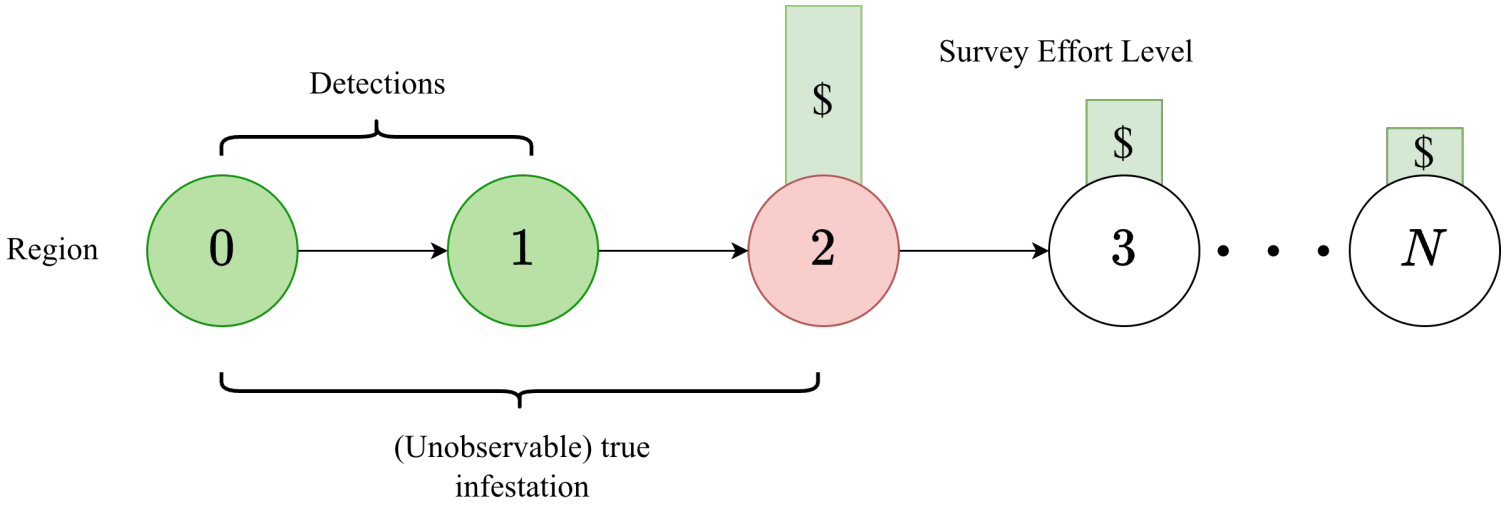
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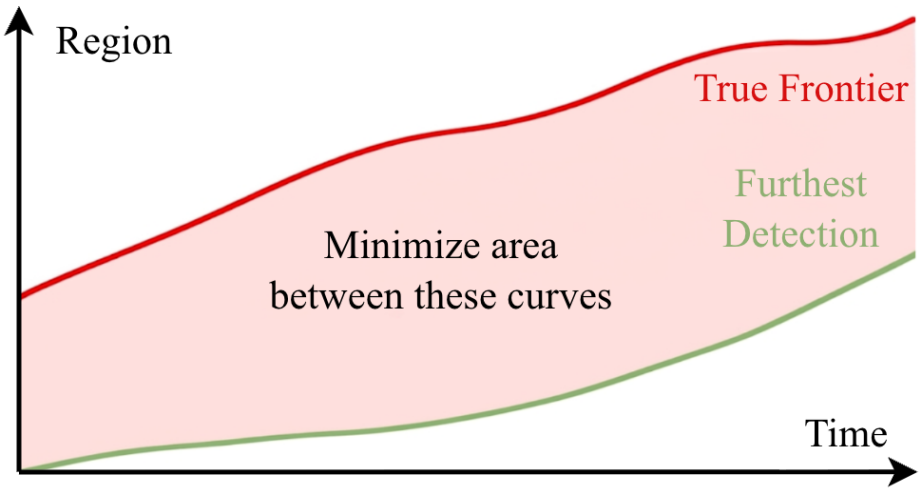
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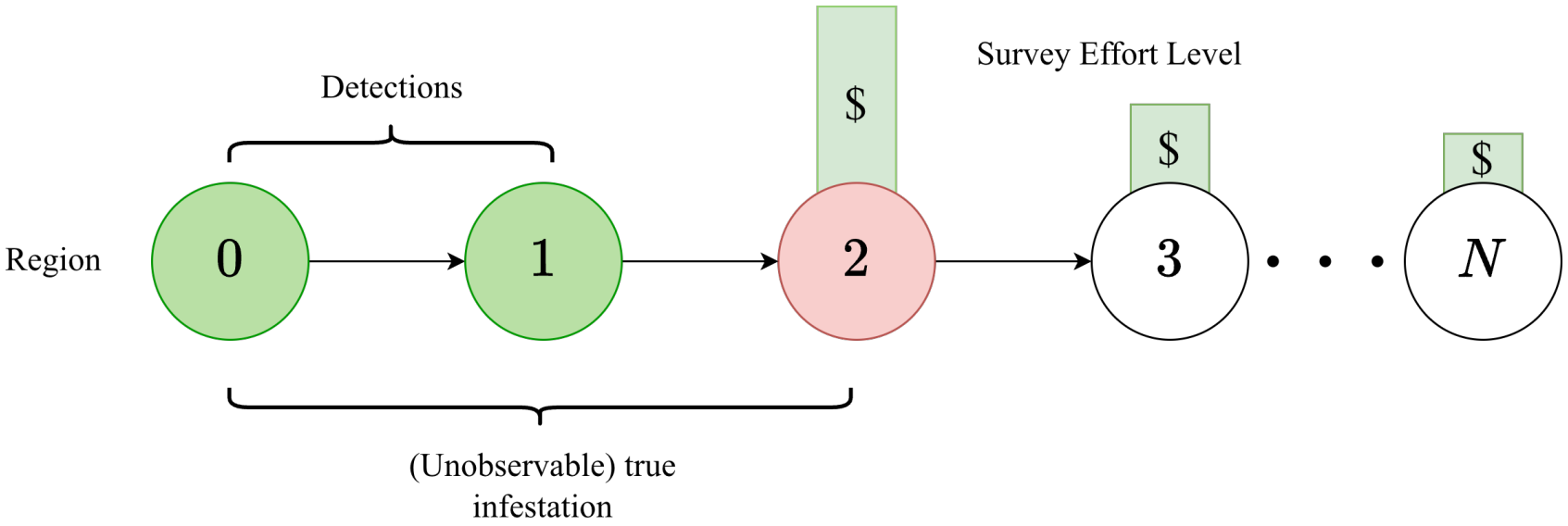


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- Formally, we want to minimize the area of undetected infestation.

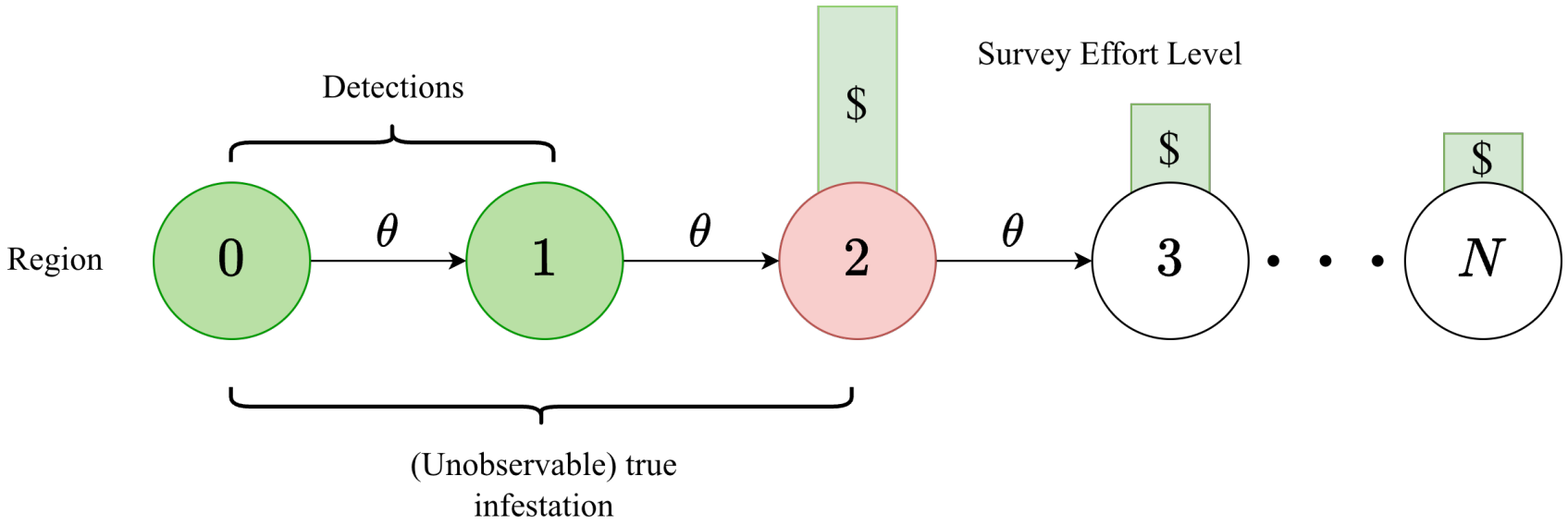


Model Parameters



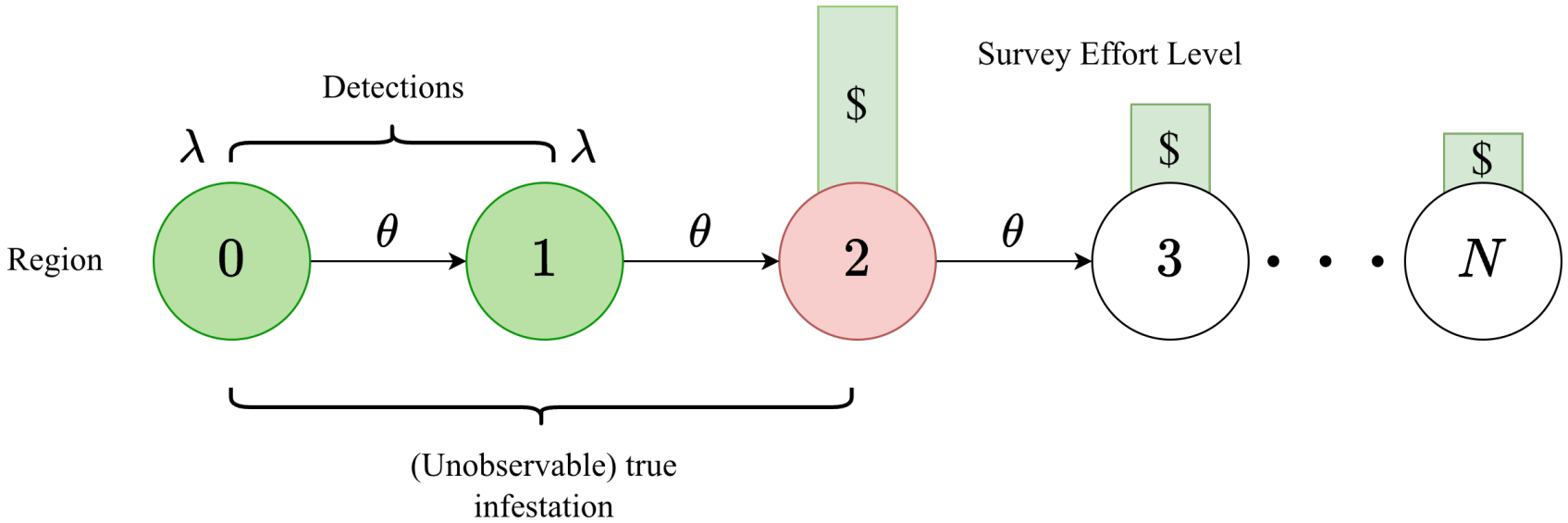
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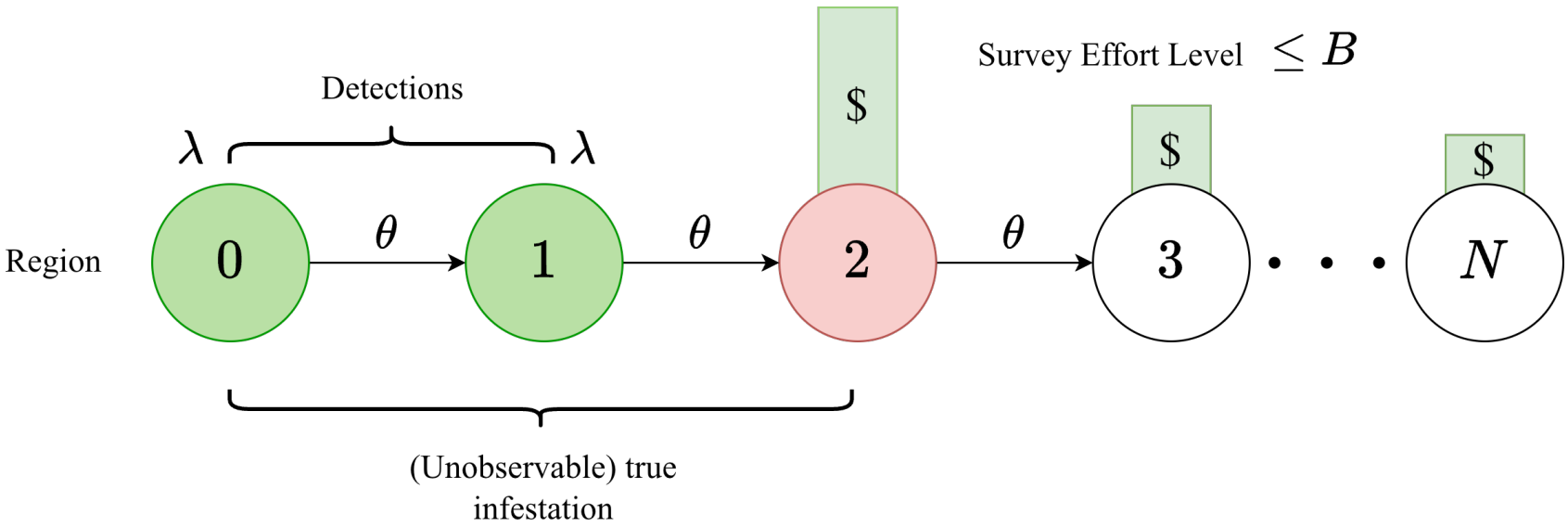
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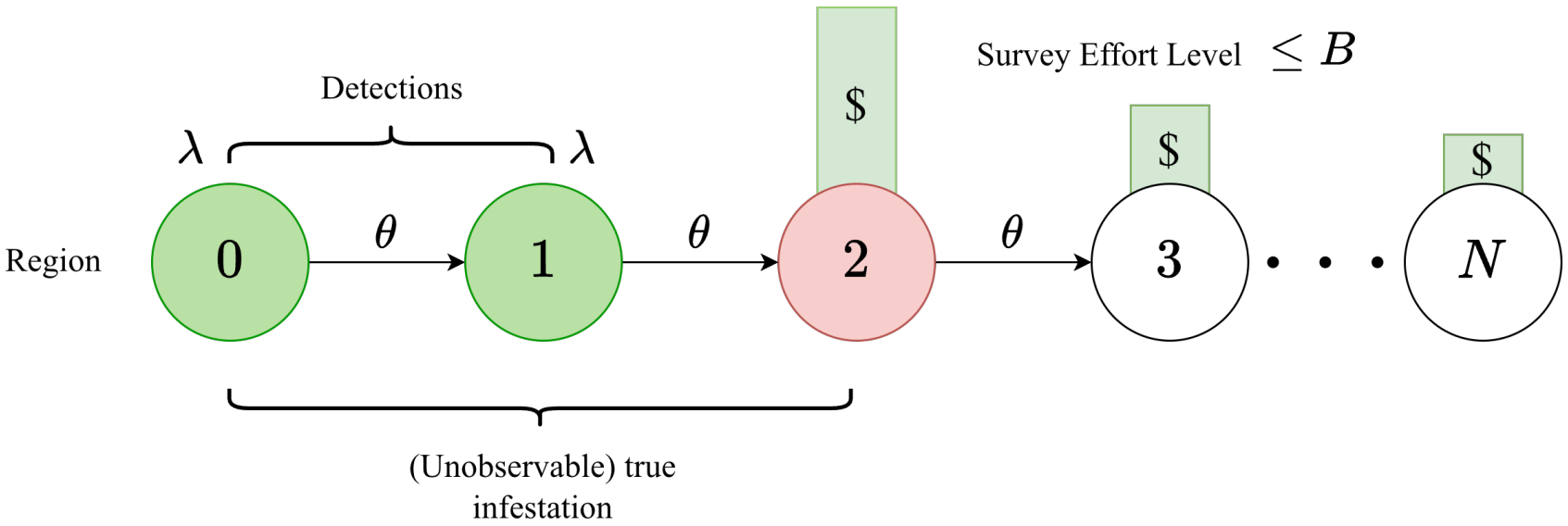
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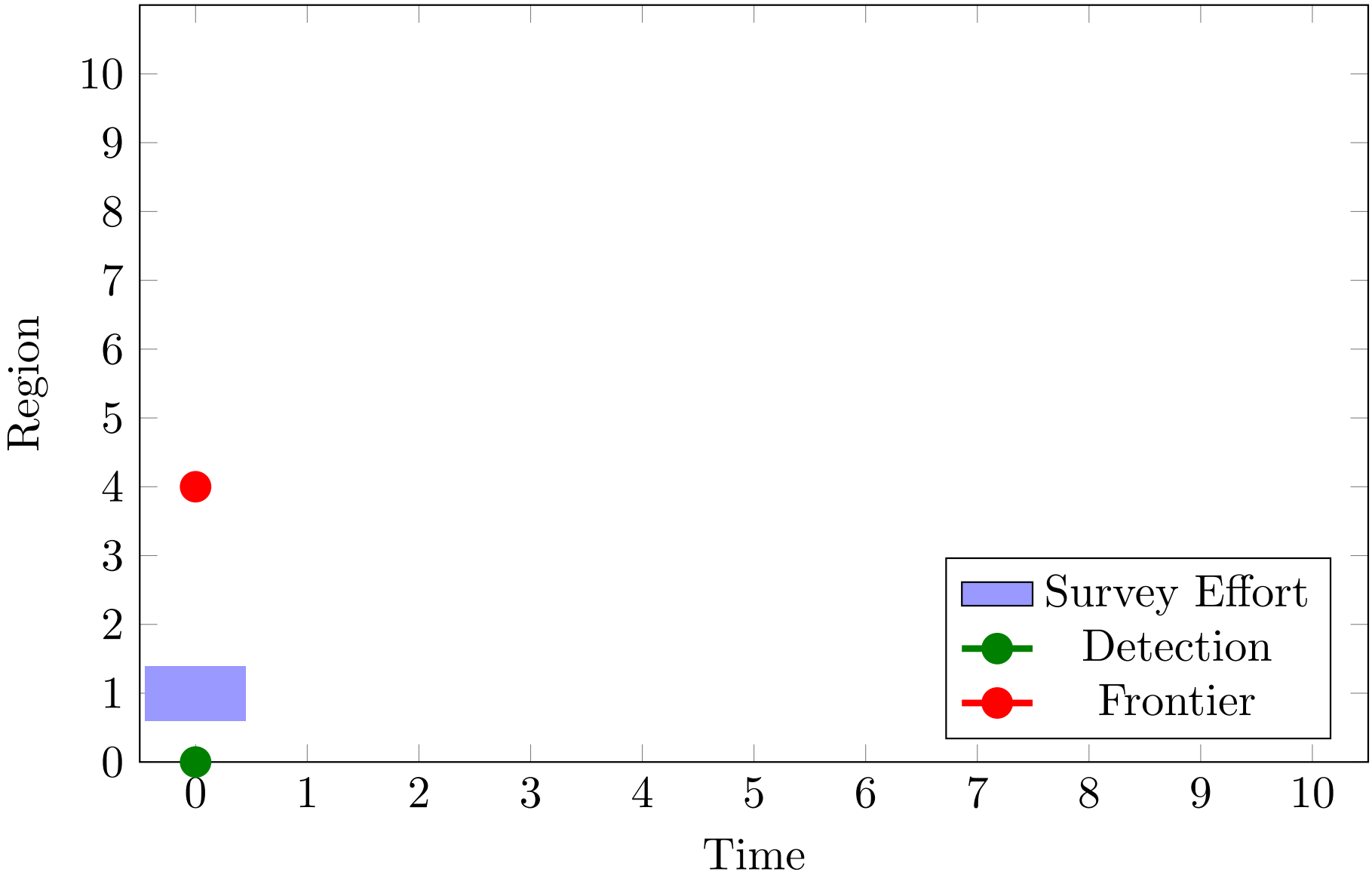


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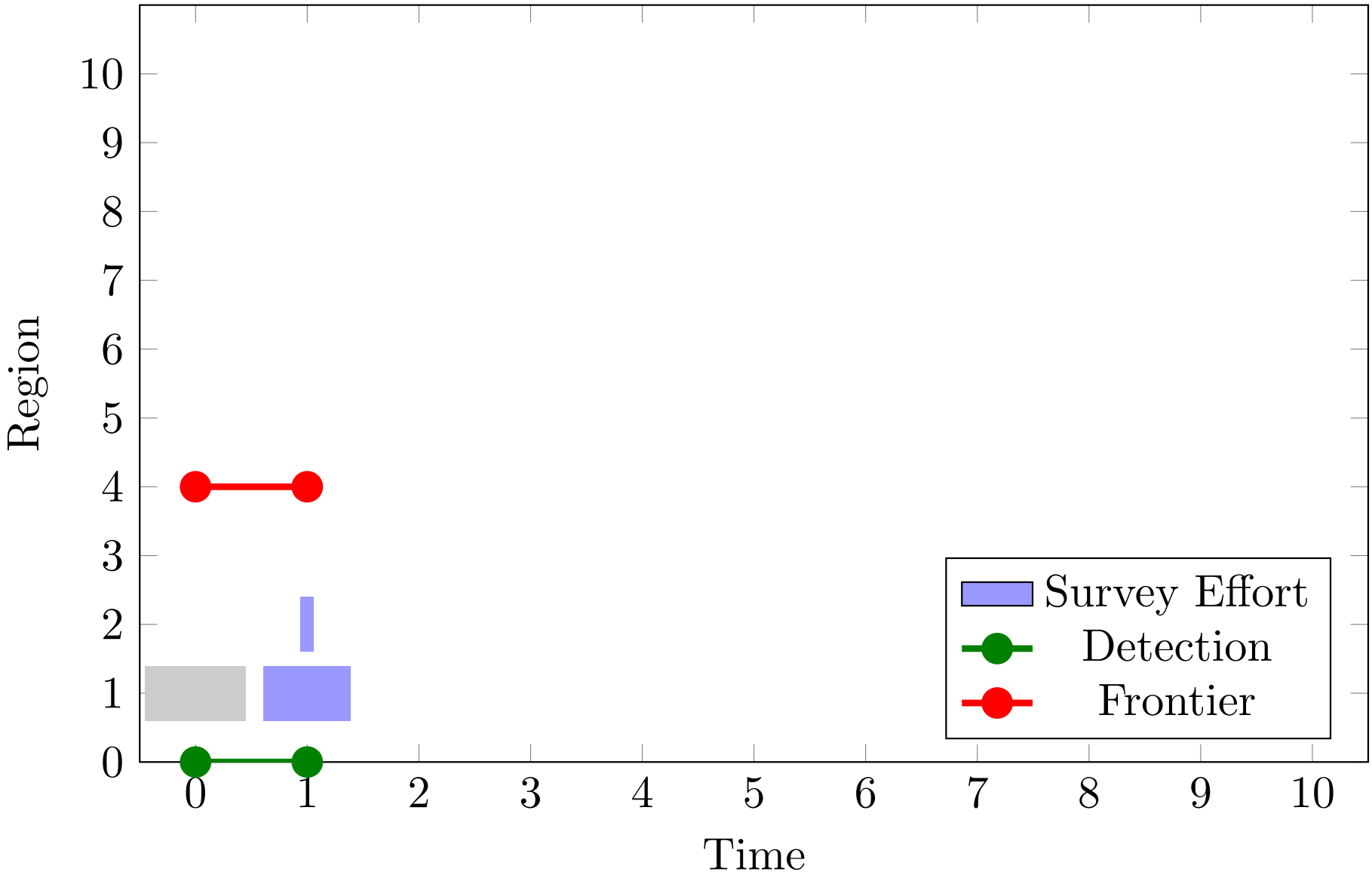
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- **Prior distribution (belief):** A probability distribution over the regions representing where you initially think the frontier is



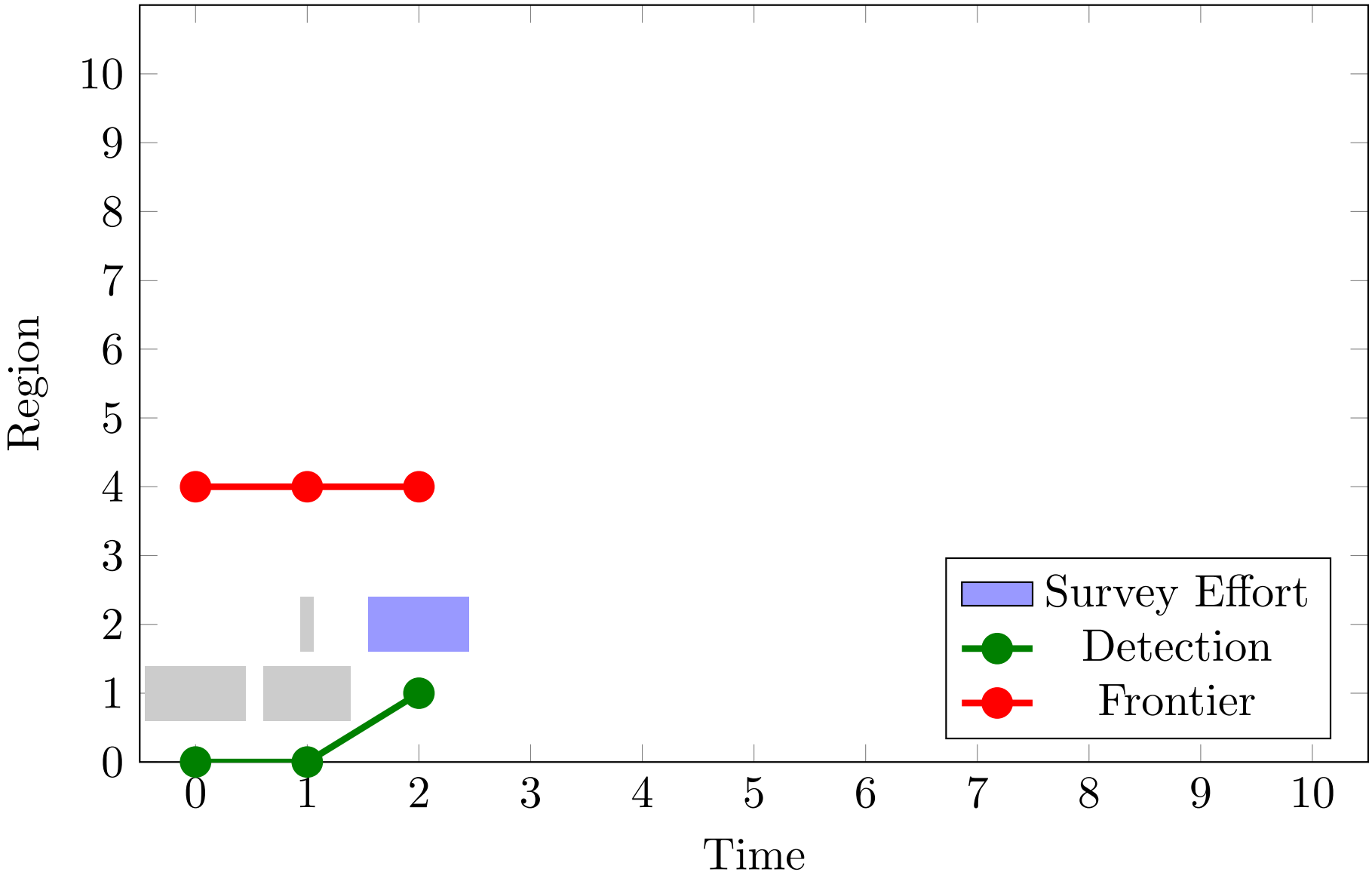
Example Invasion



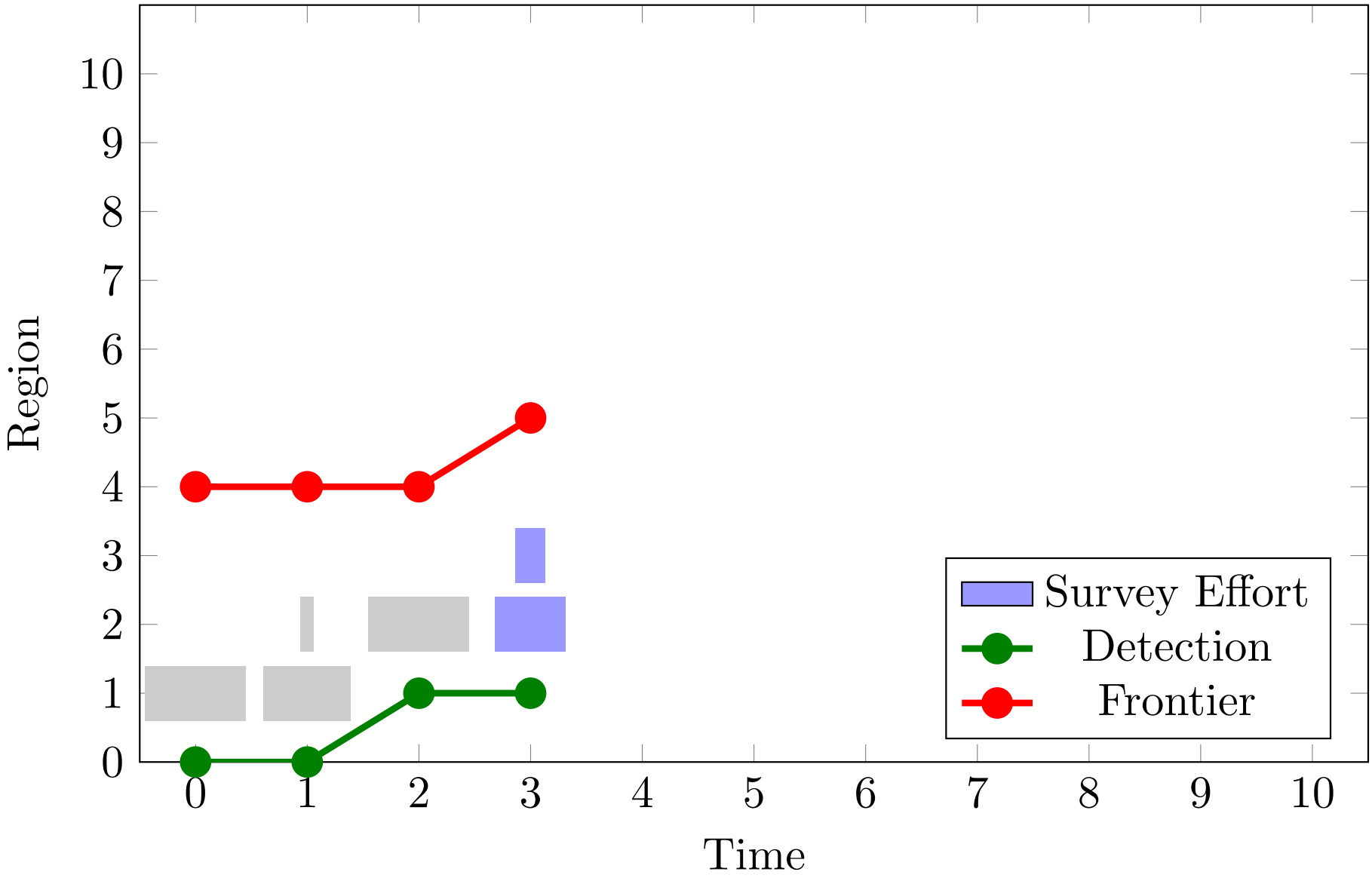
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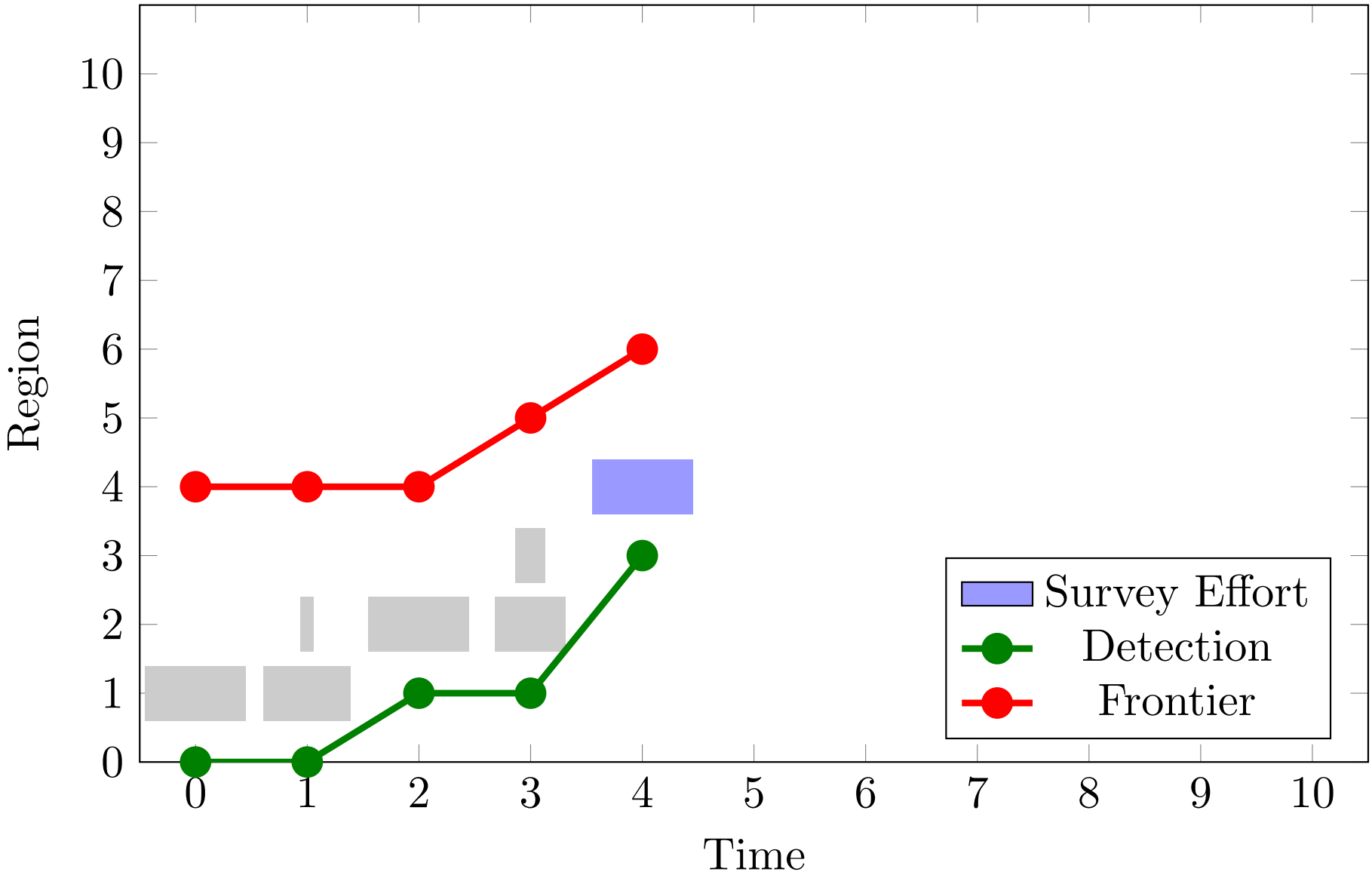
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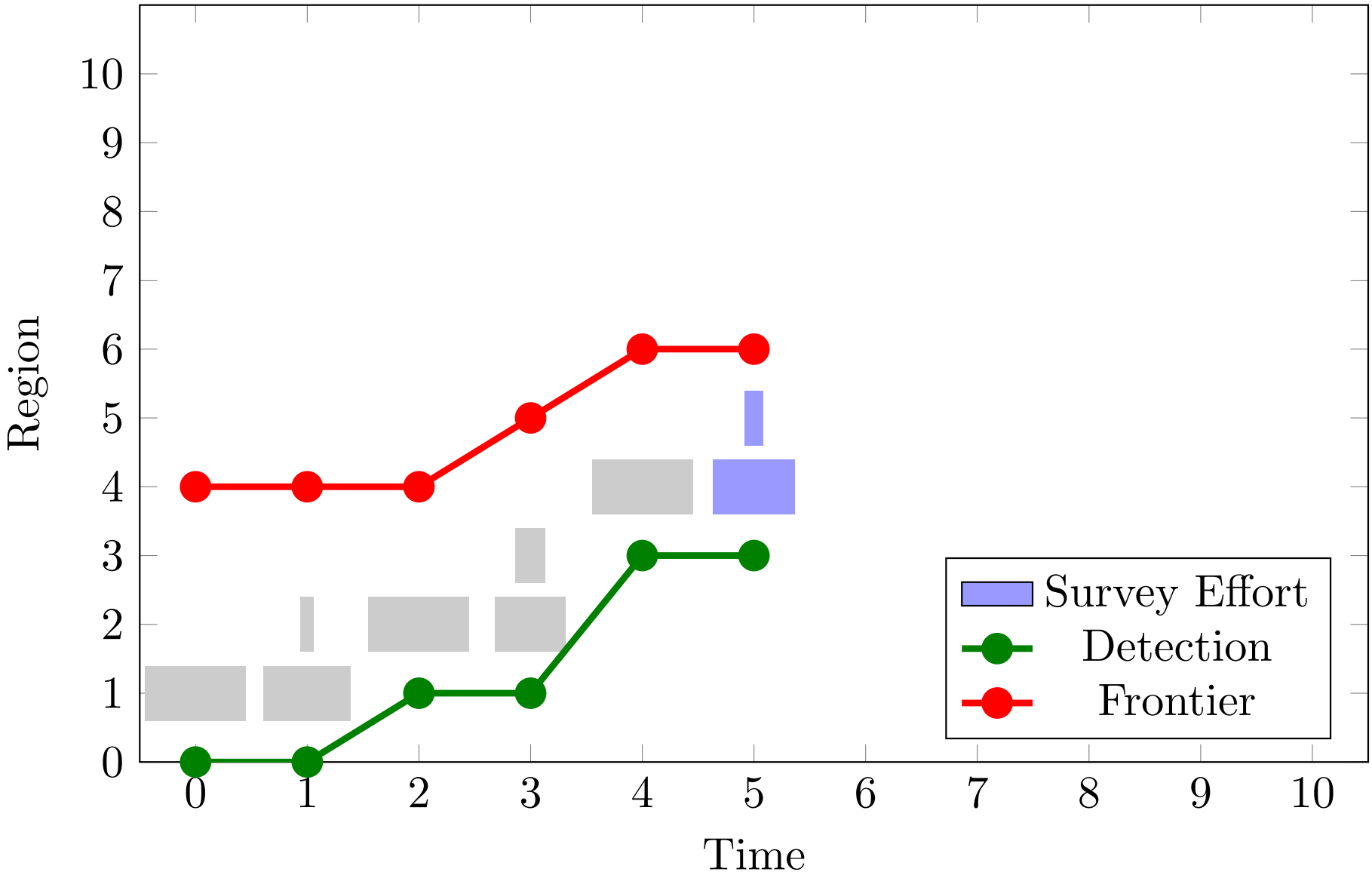
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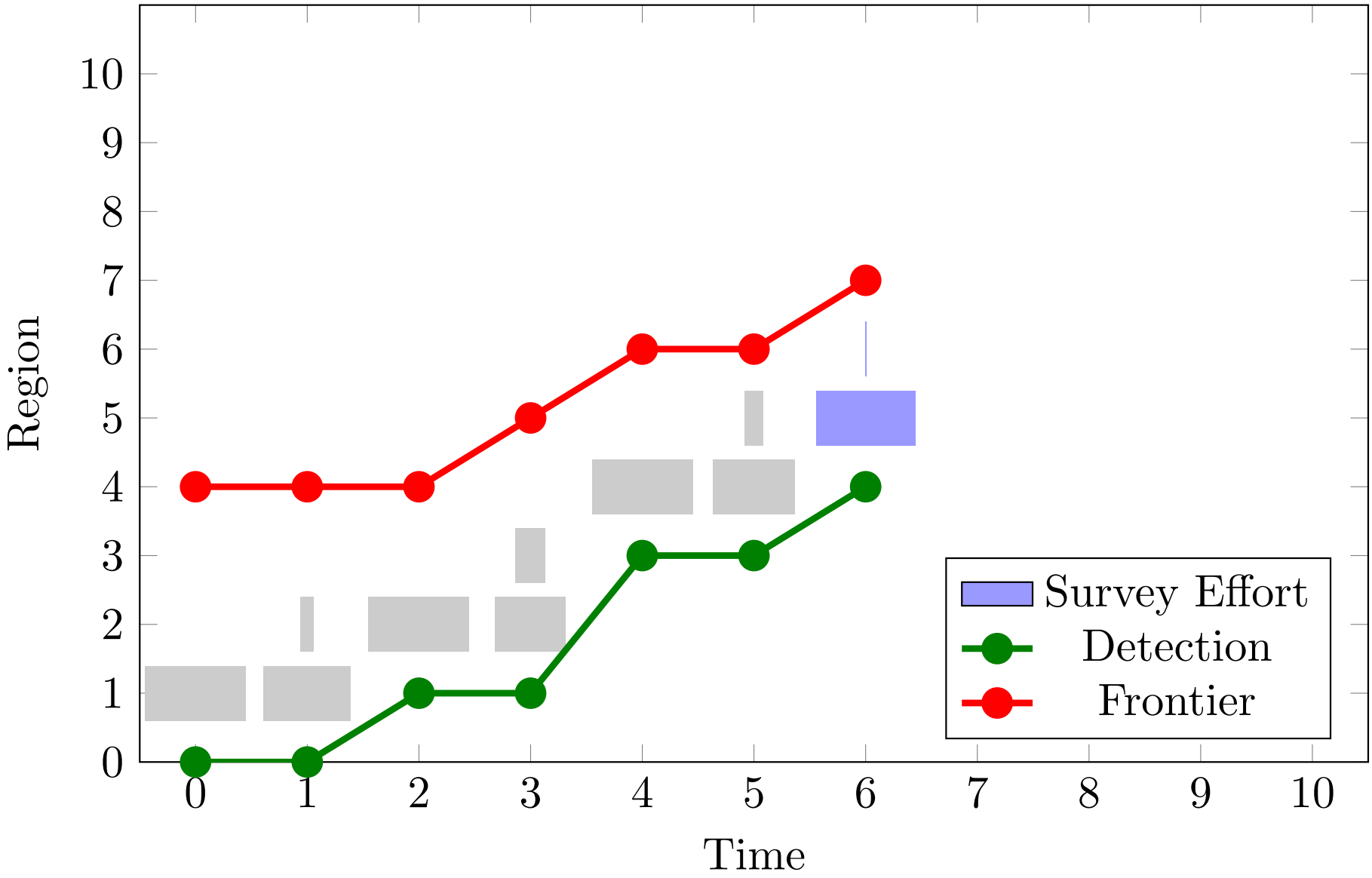
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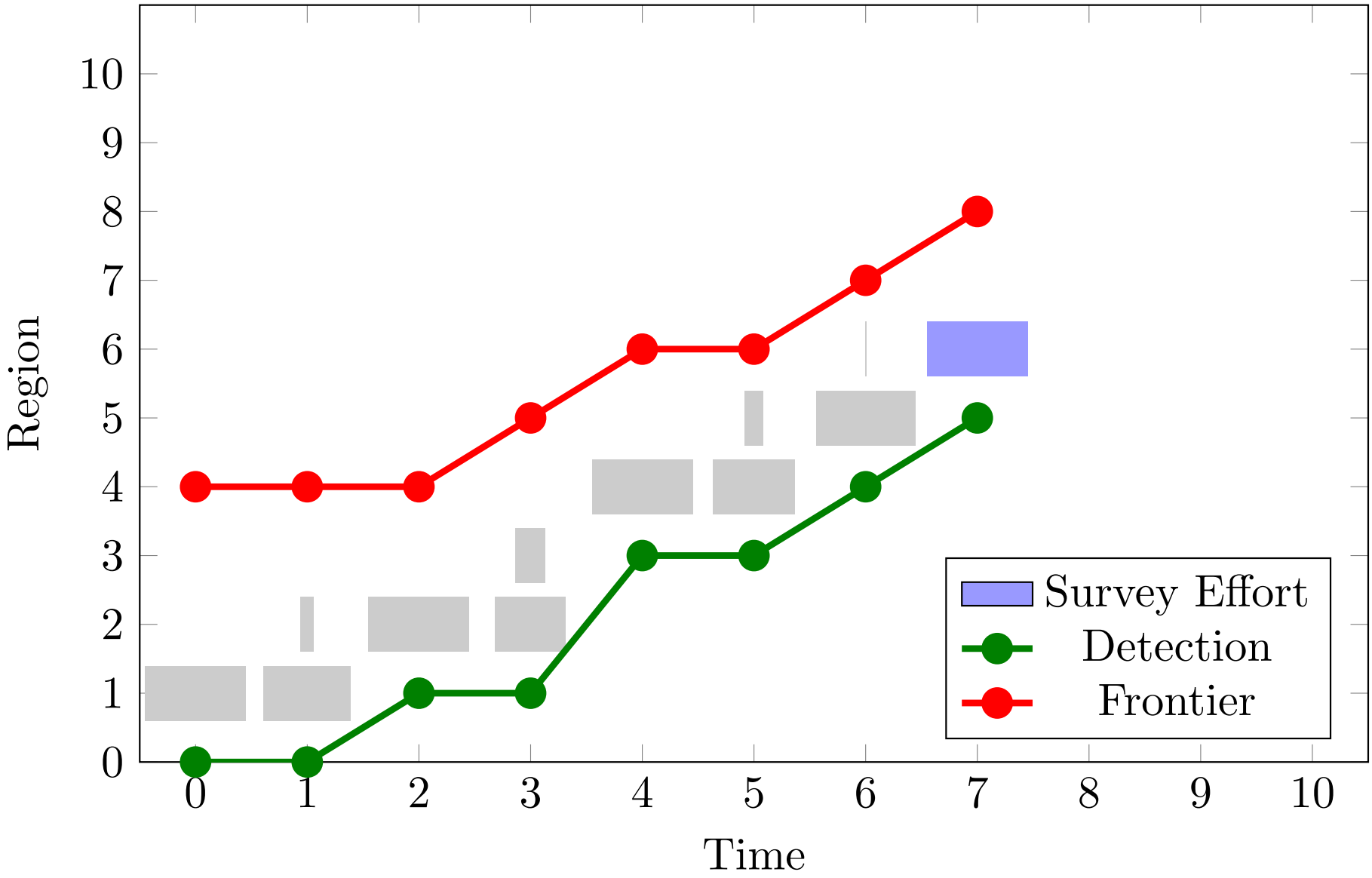
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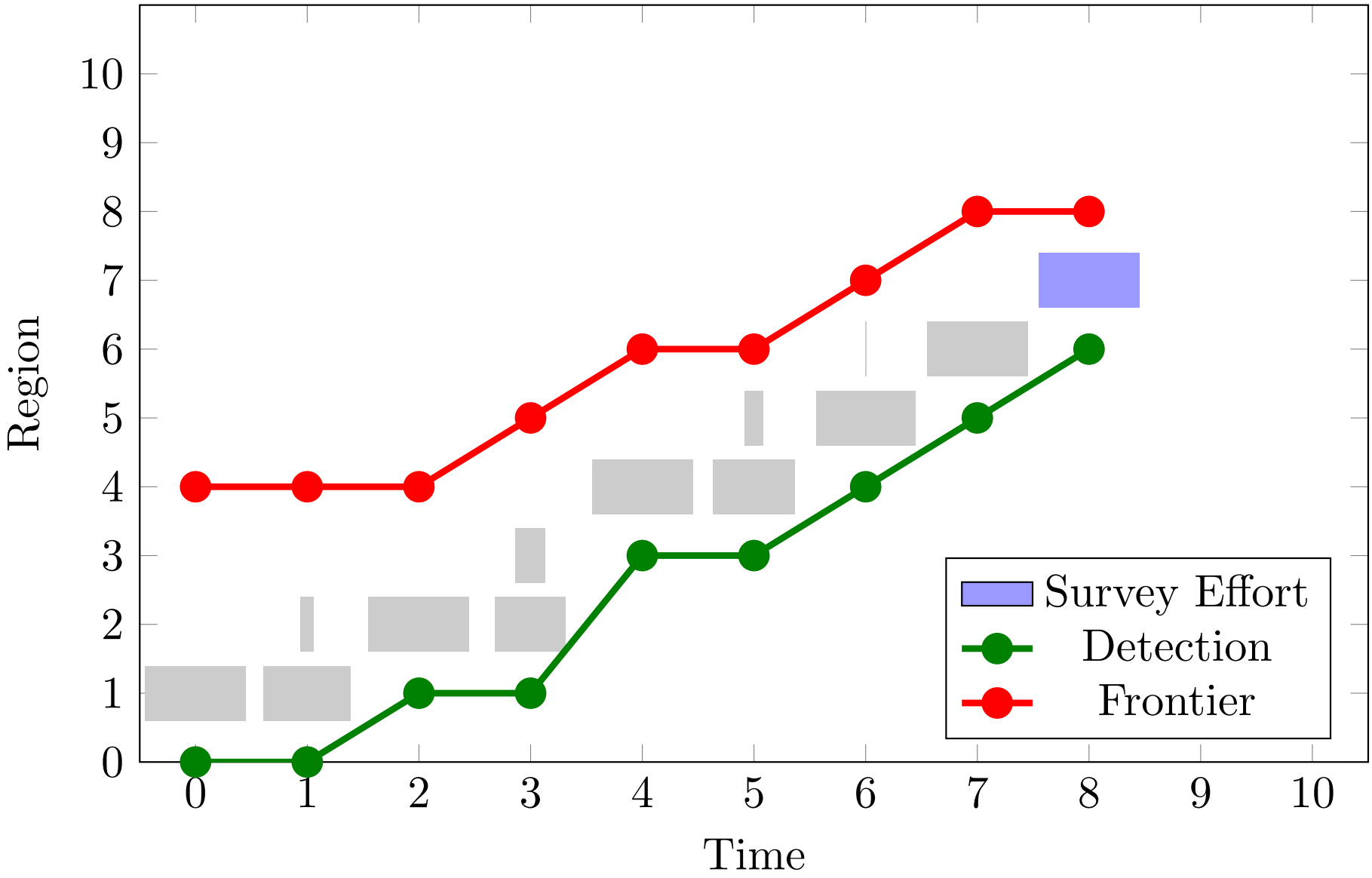
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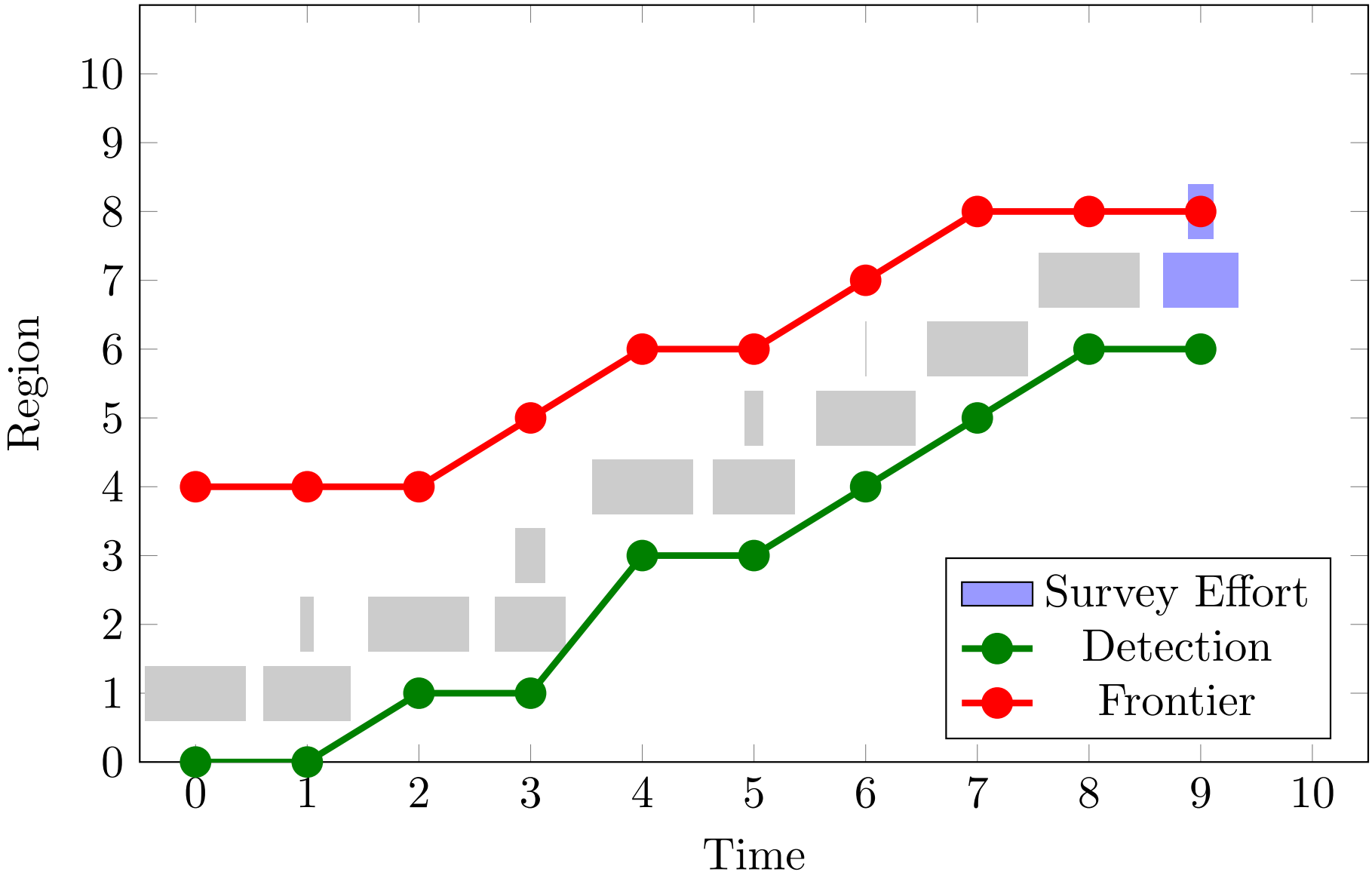
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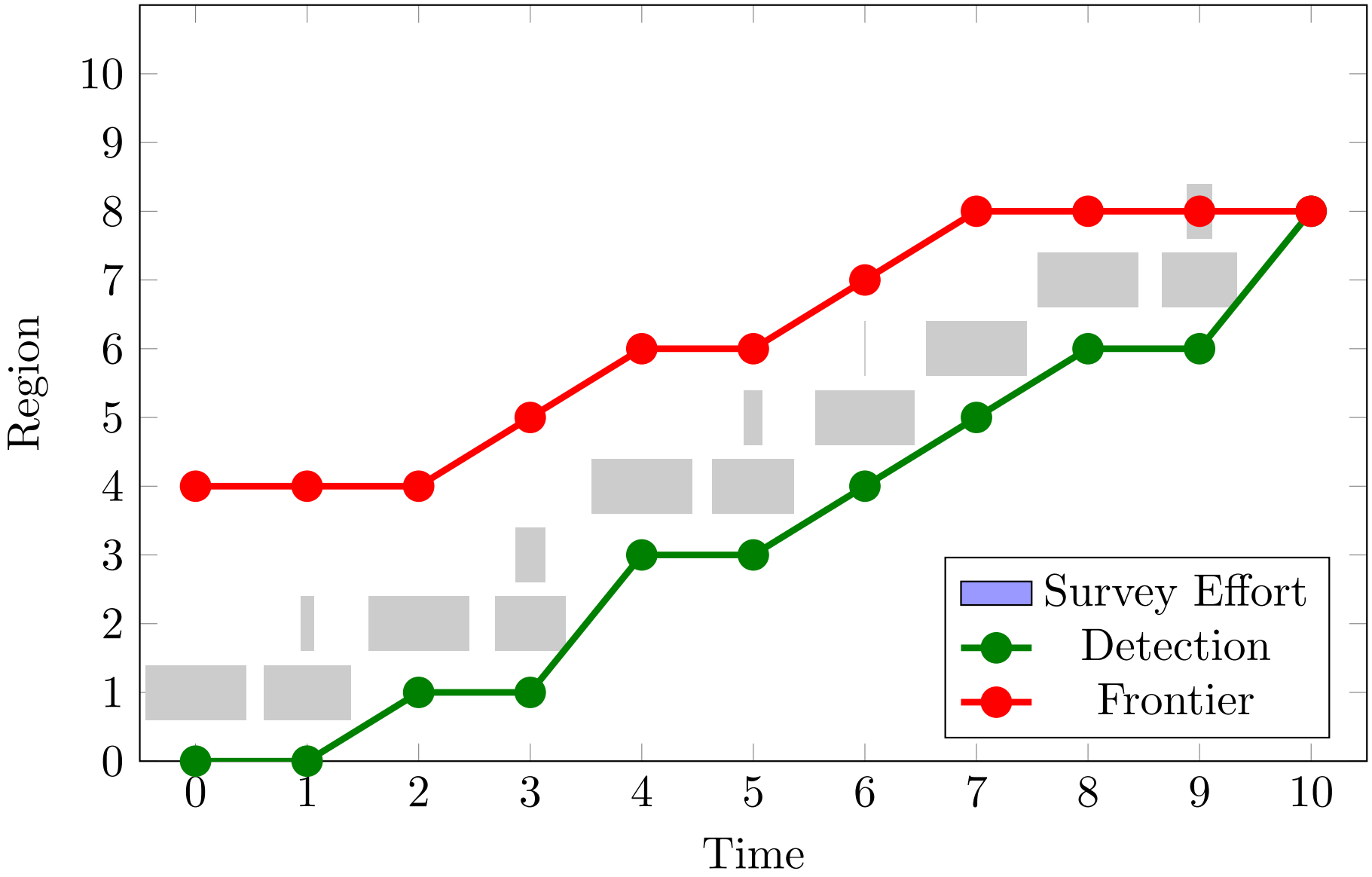
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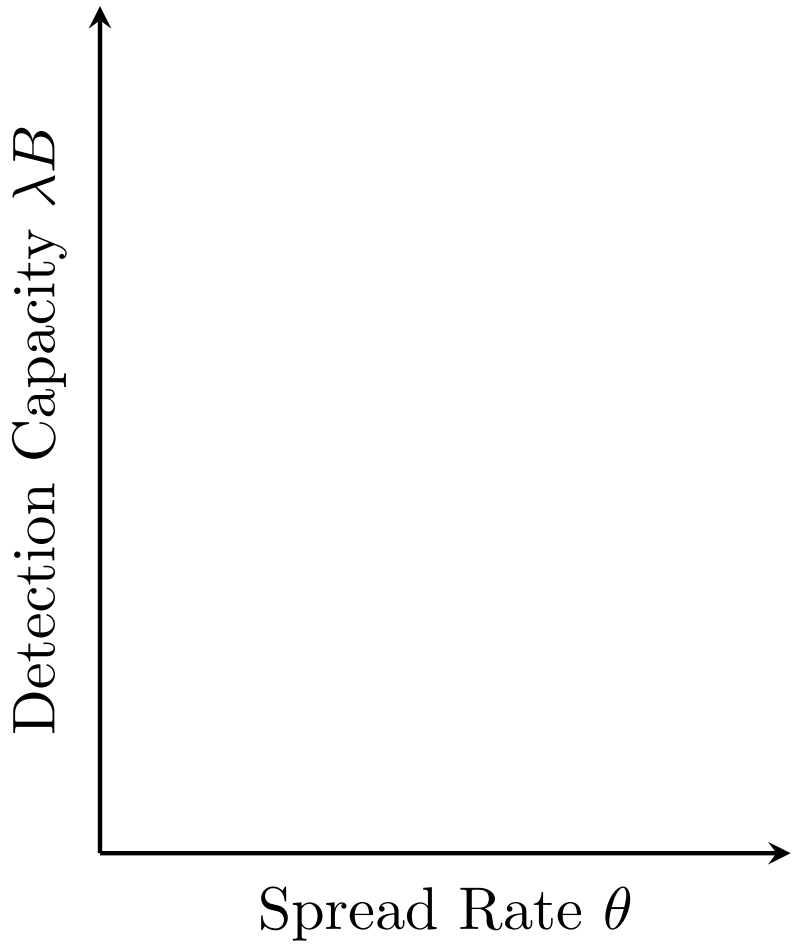
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- How can we **learn**, from both detections and lack of detections?

Assumptions

- The problem is one-dimensional
- The pest moves according to a travelling wave: to get to one region it must go through the previous ones
- The spread rate and detection probability can be approximated from historical observations and spread data
- Each region can only be either infested or uninfested
- Currently, we assume delimiting surveys only: no prevention or management (we plan to include this at a later stage)

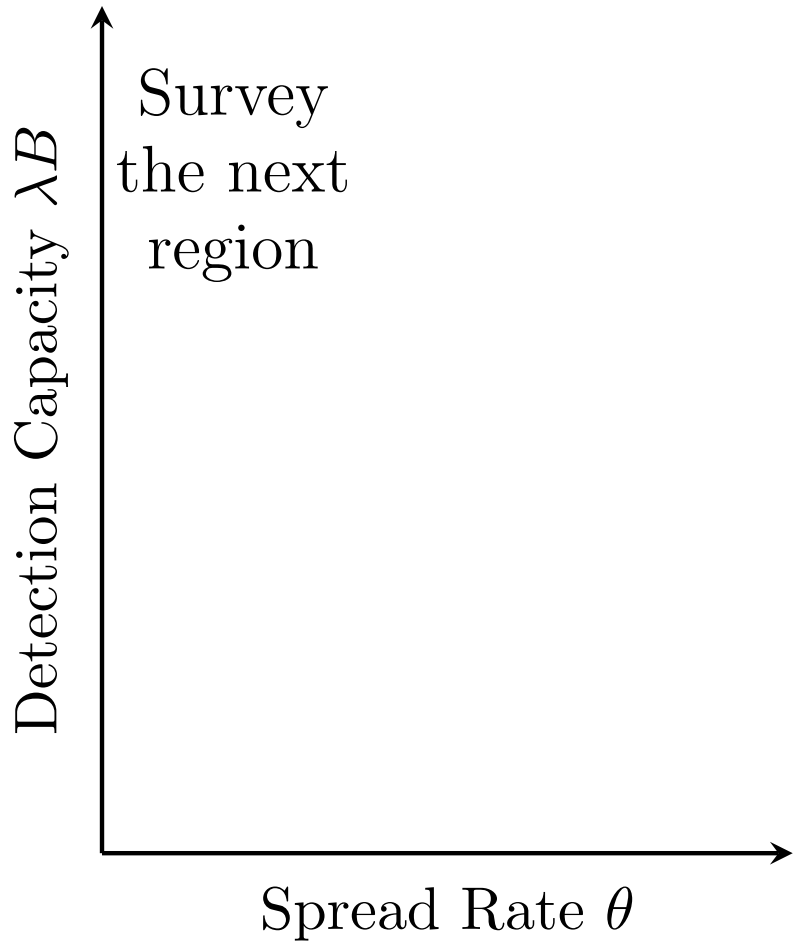
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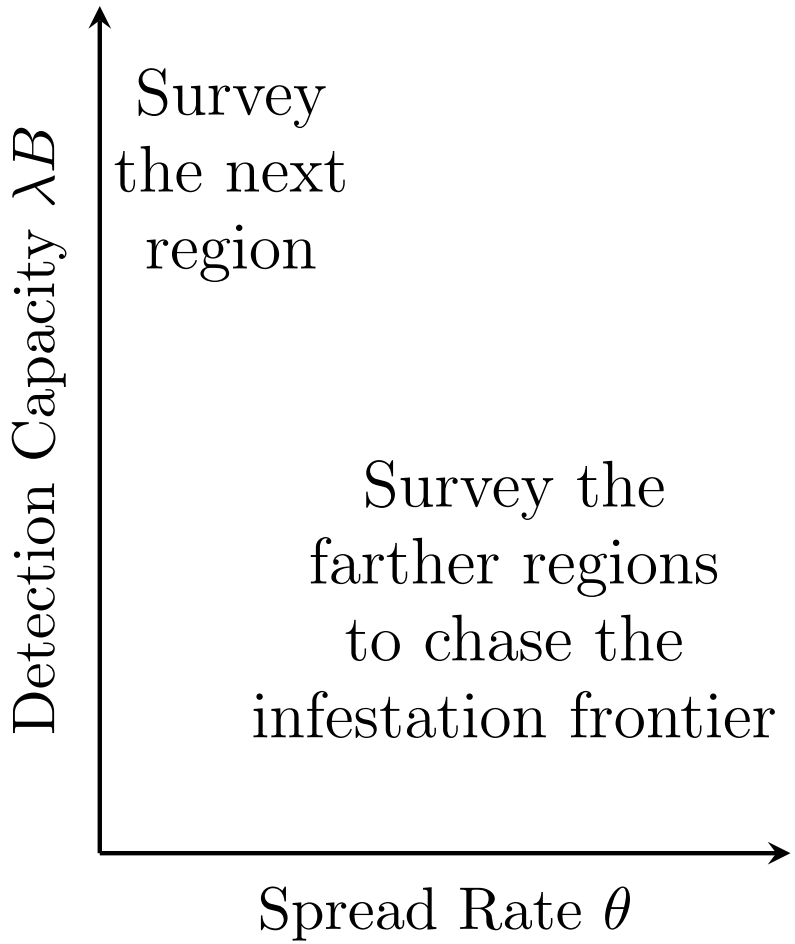
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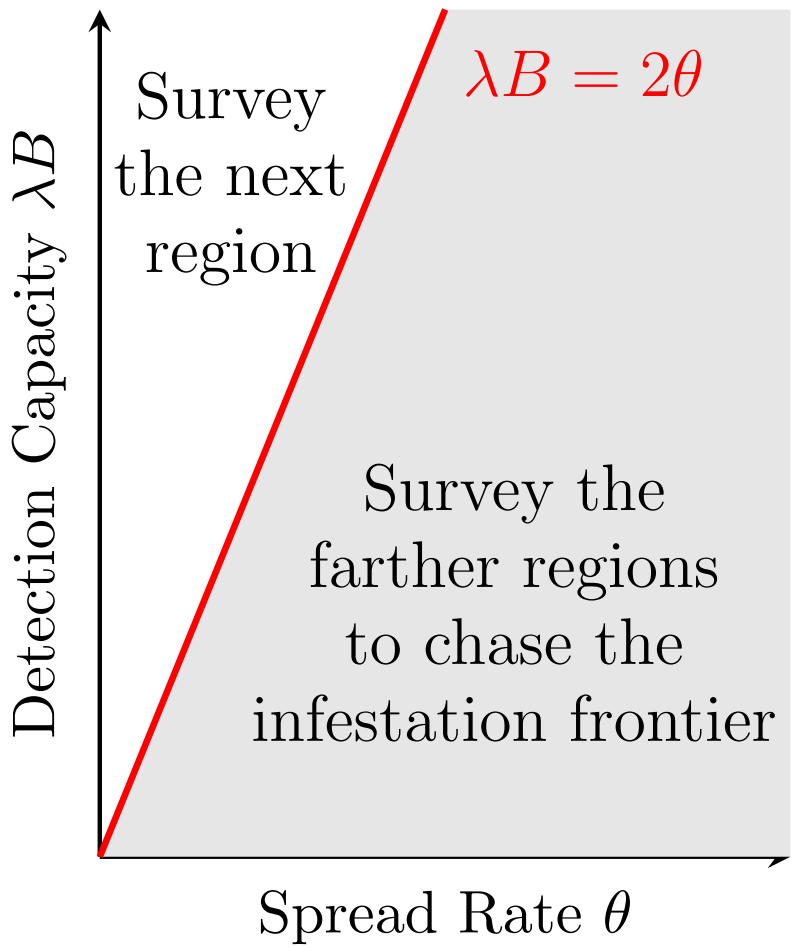
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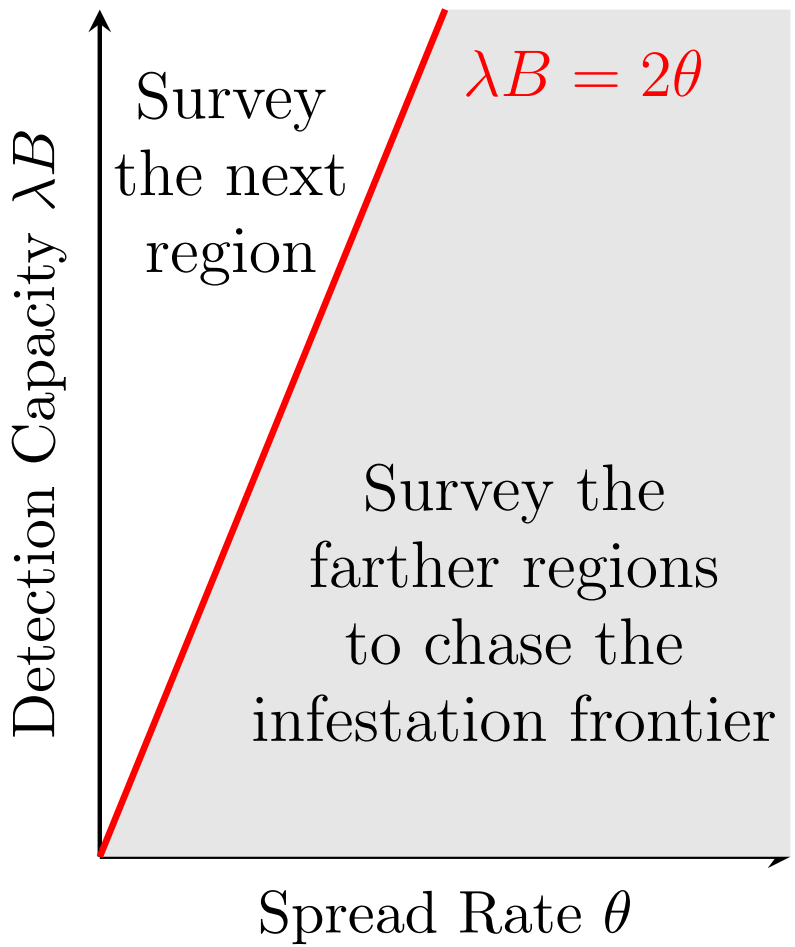
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- The importance of sufficient budget: for $B < \theta/\lambda$, you do not have enough detection power to accurately track the spread.



How to Chase the Frontier

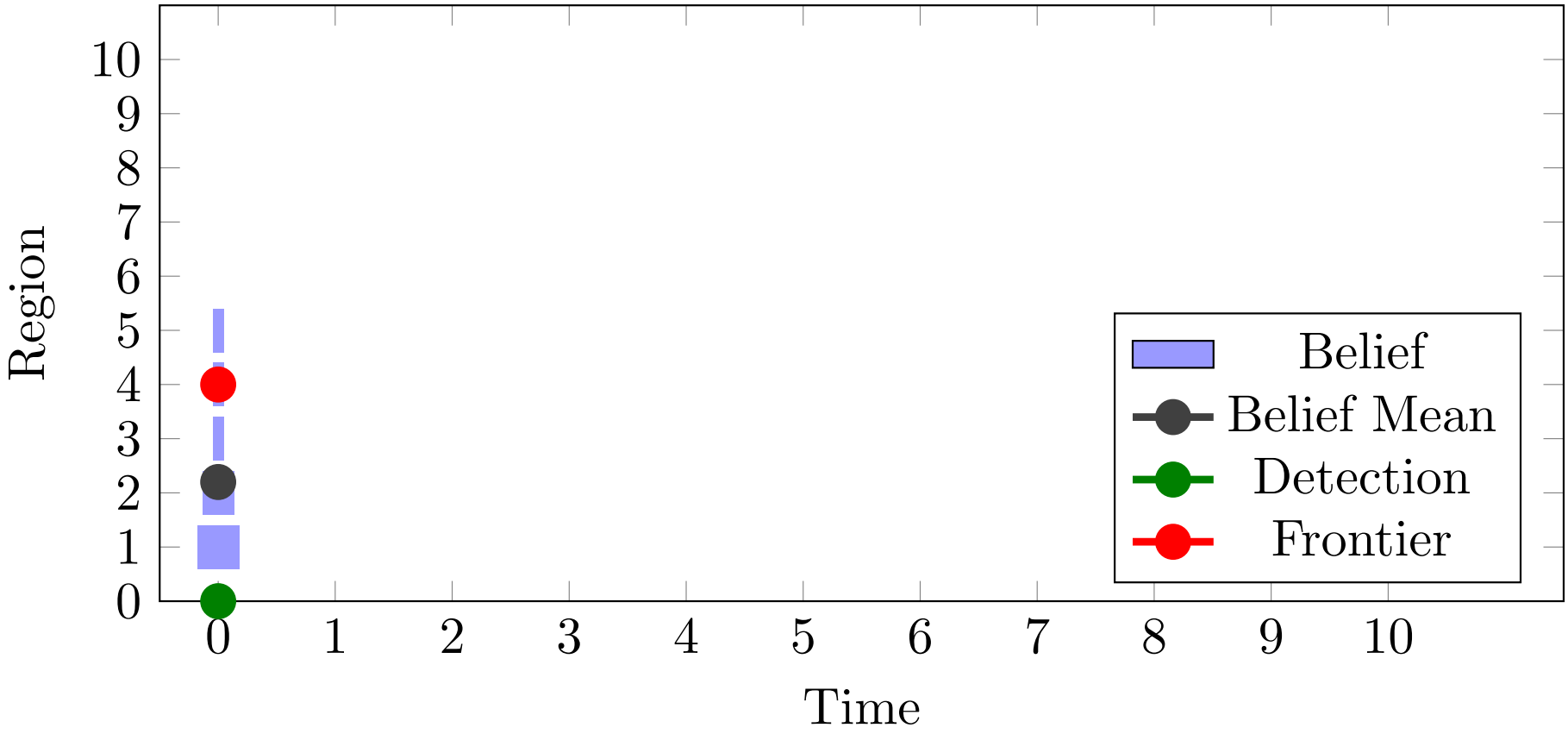
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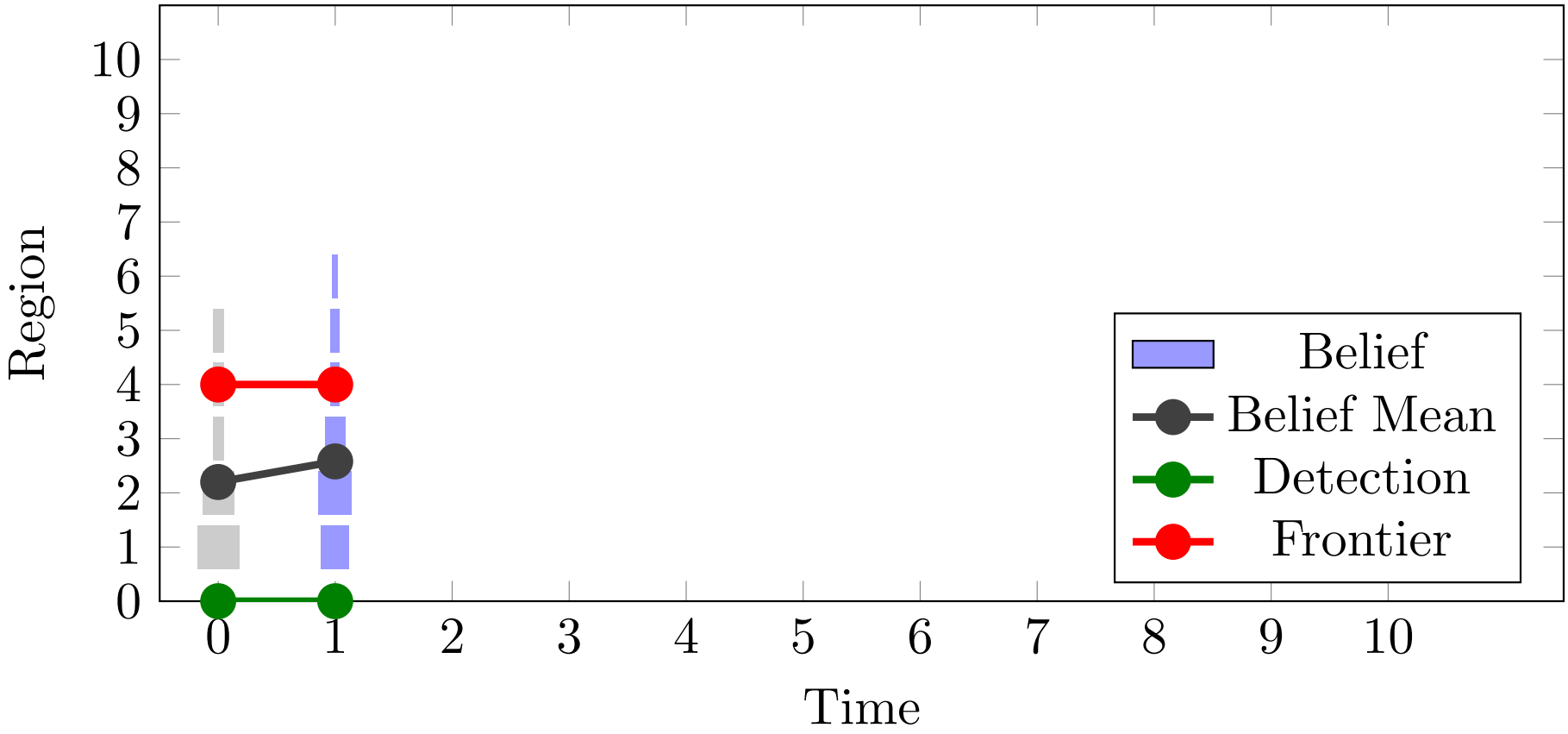
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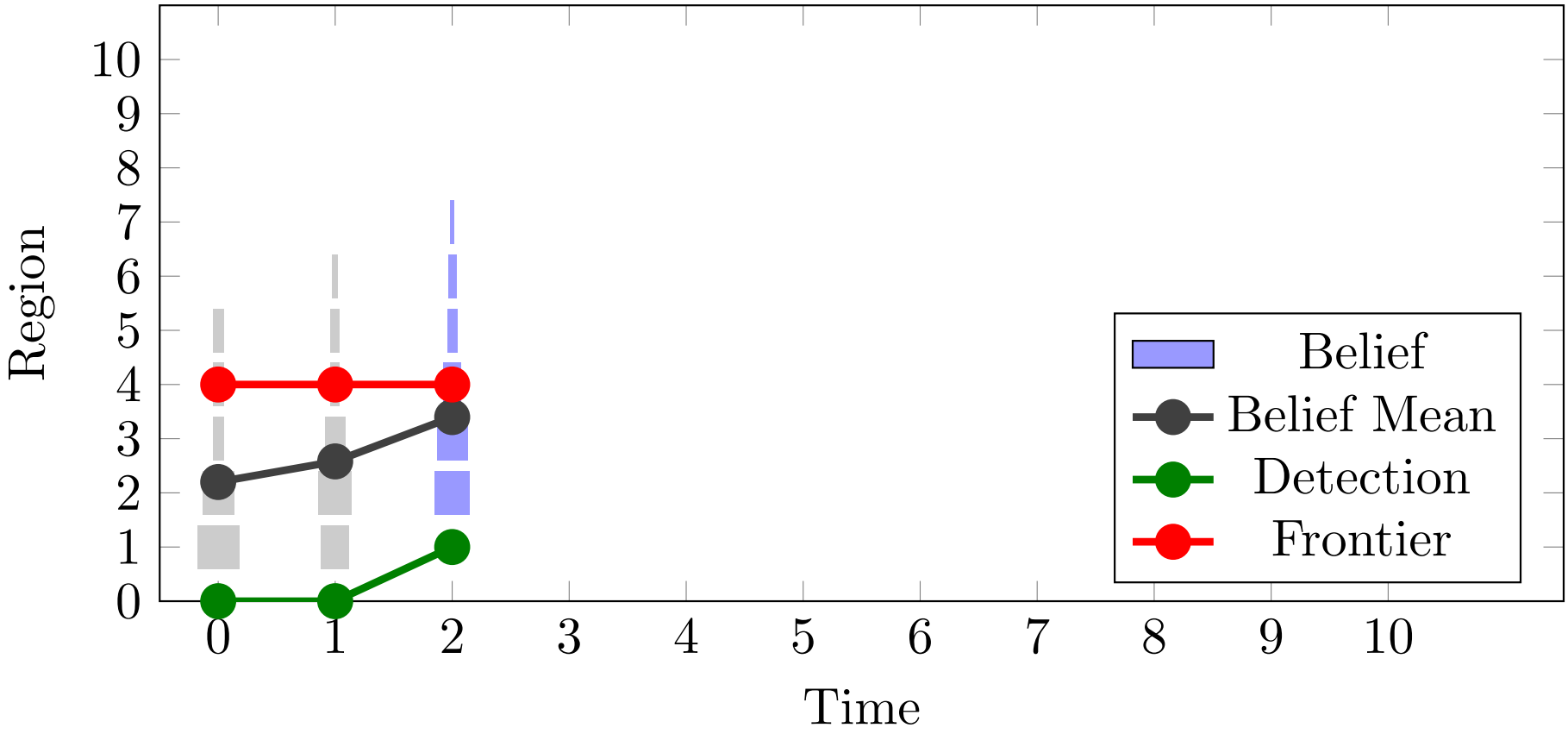
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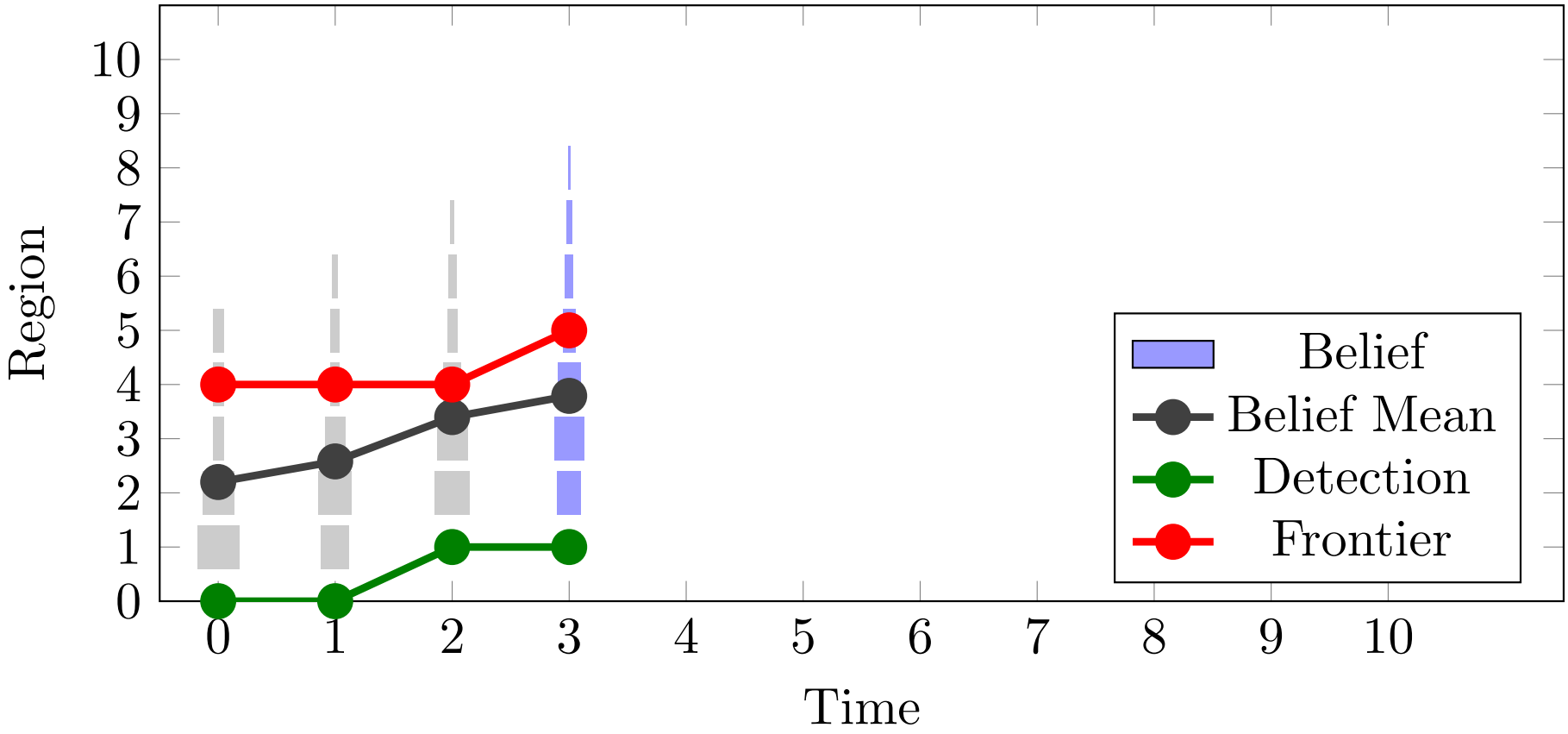
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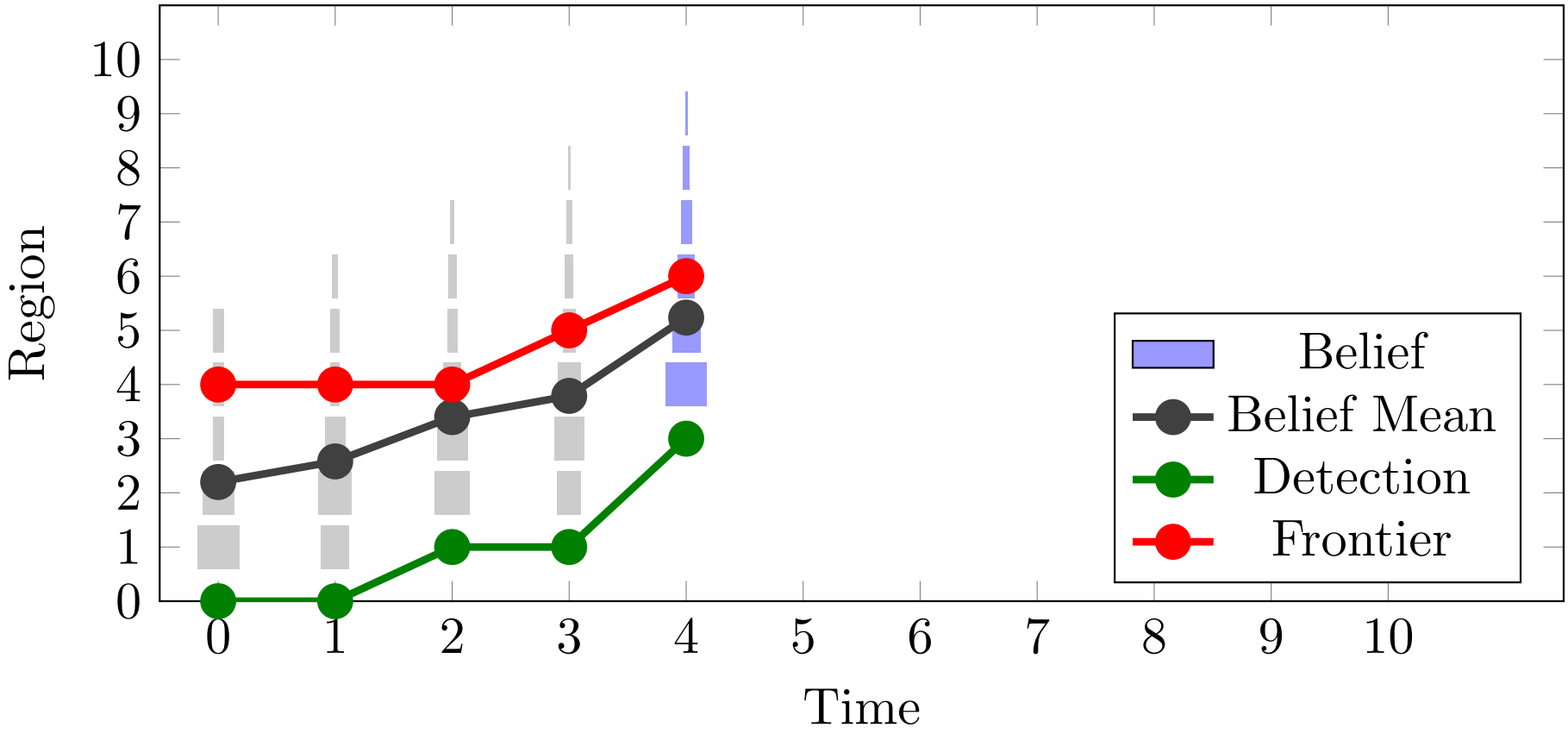
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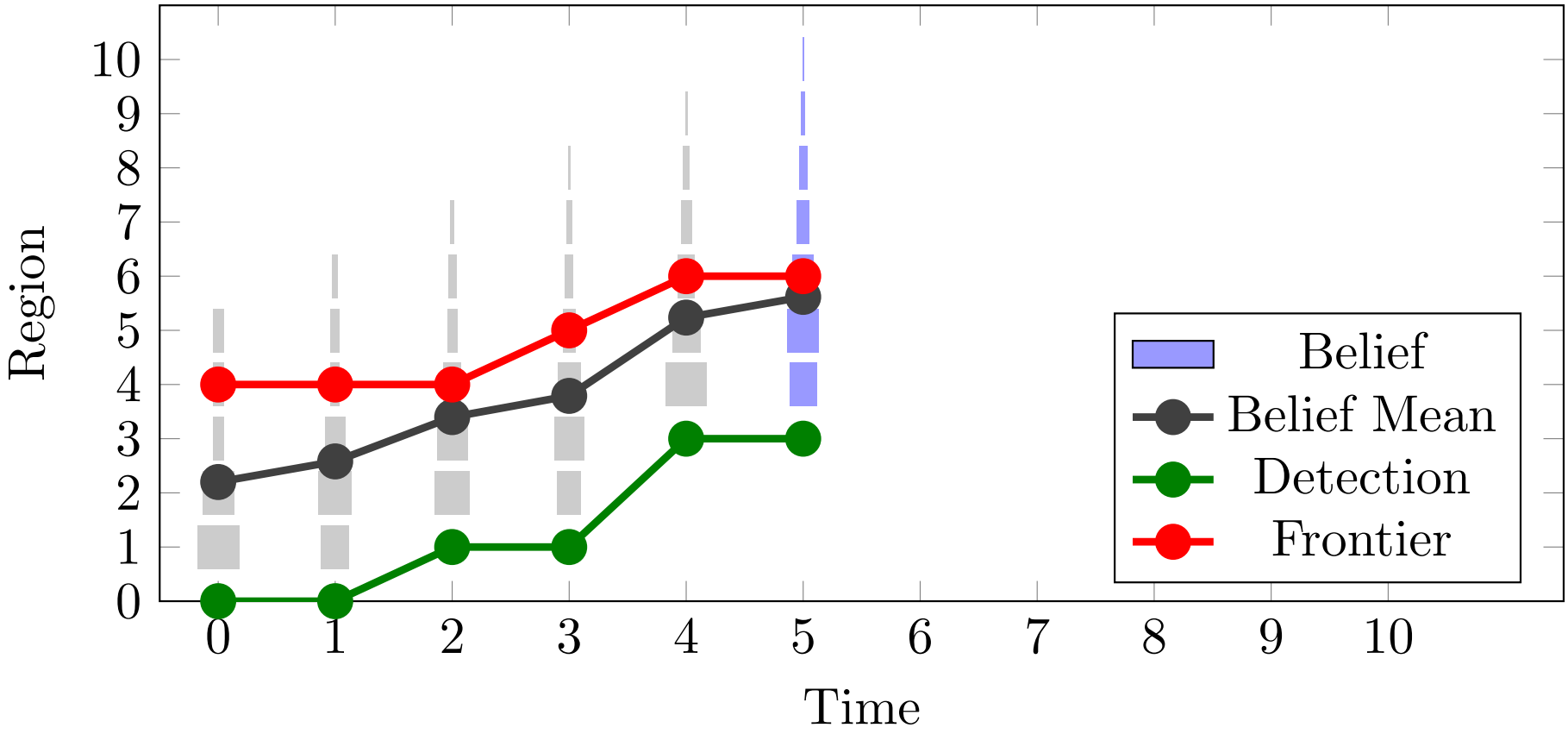
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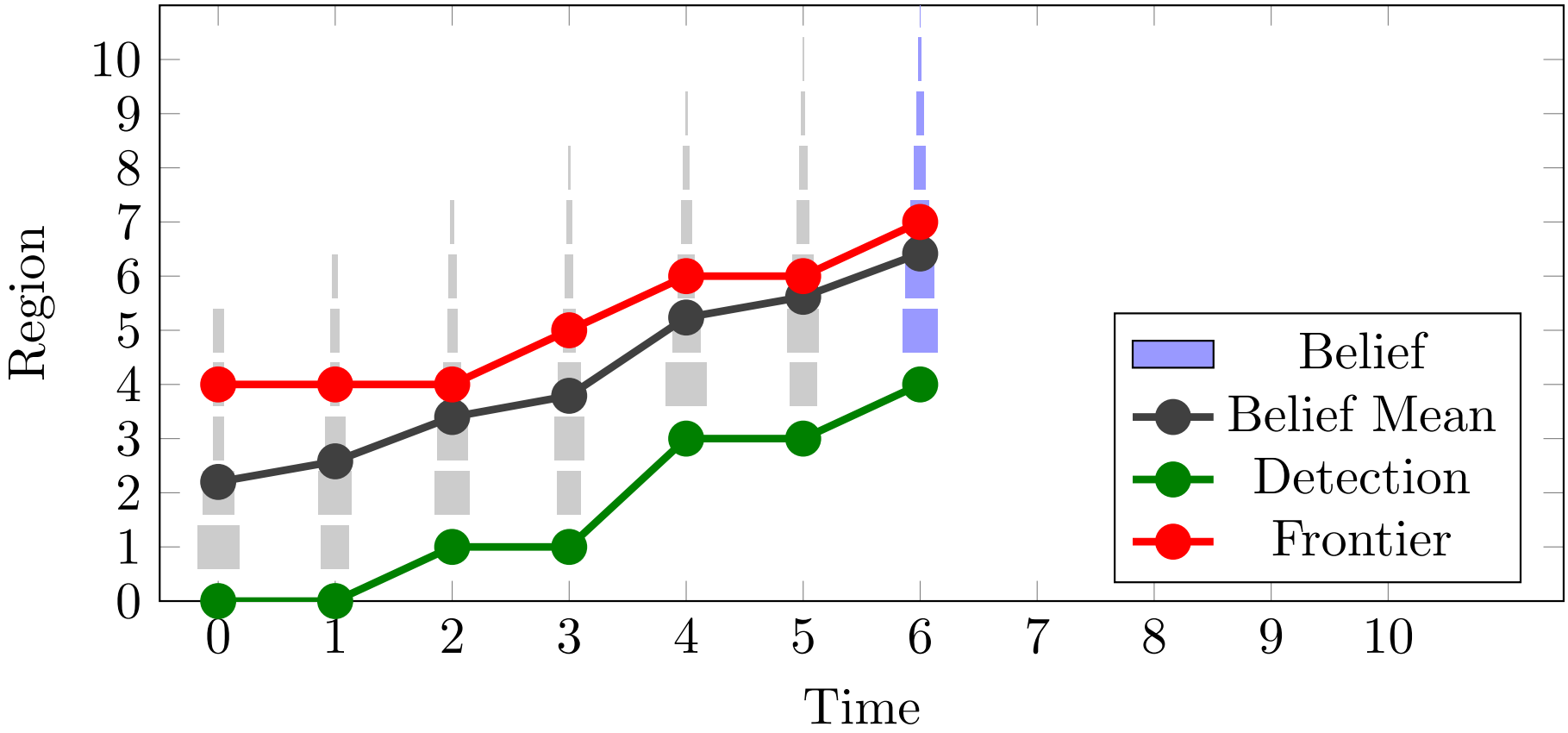
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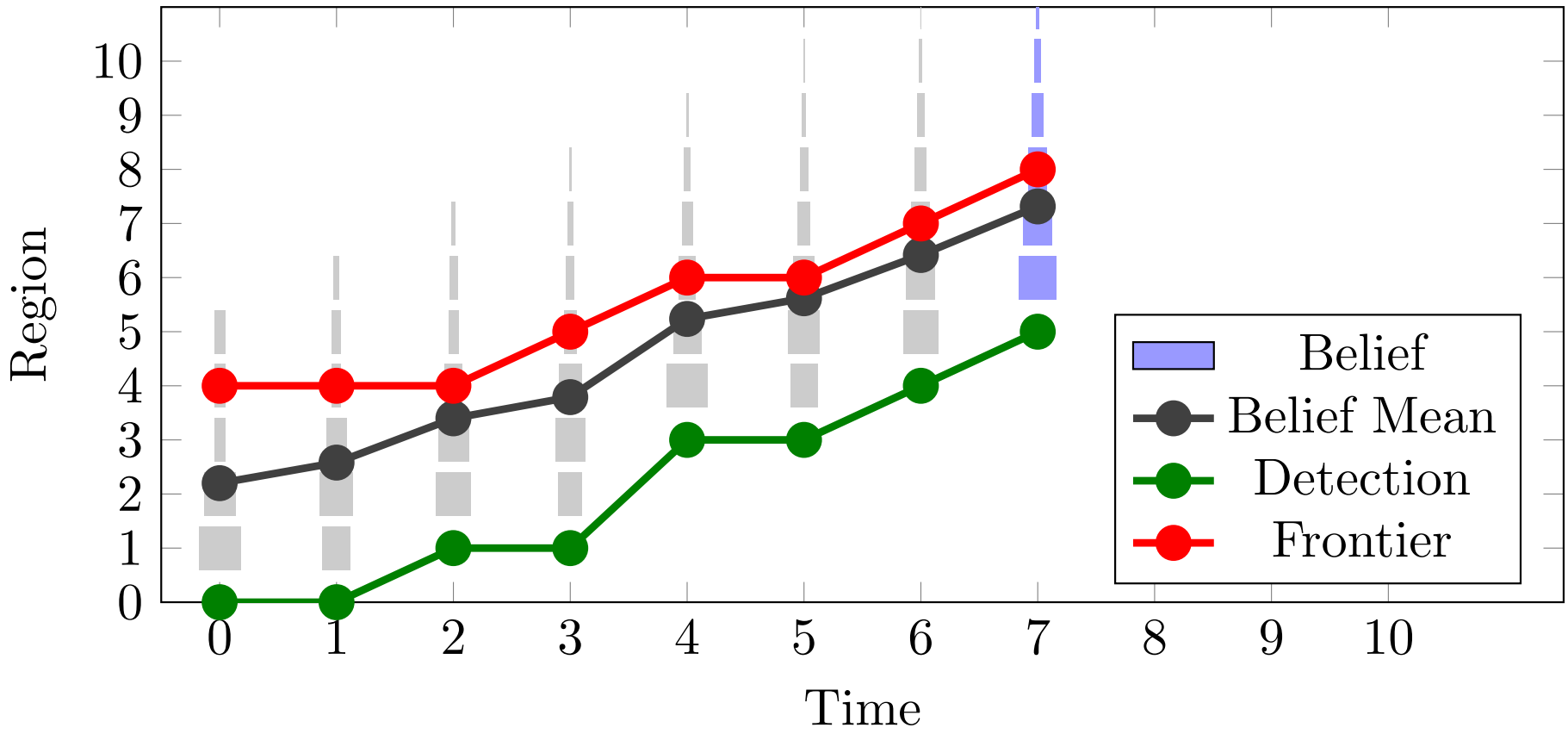
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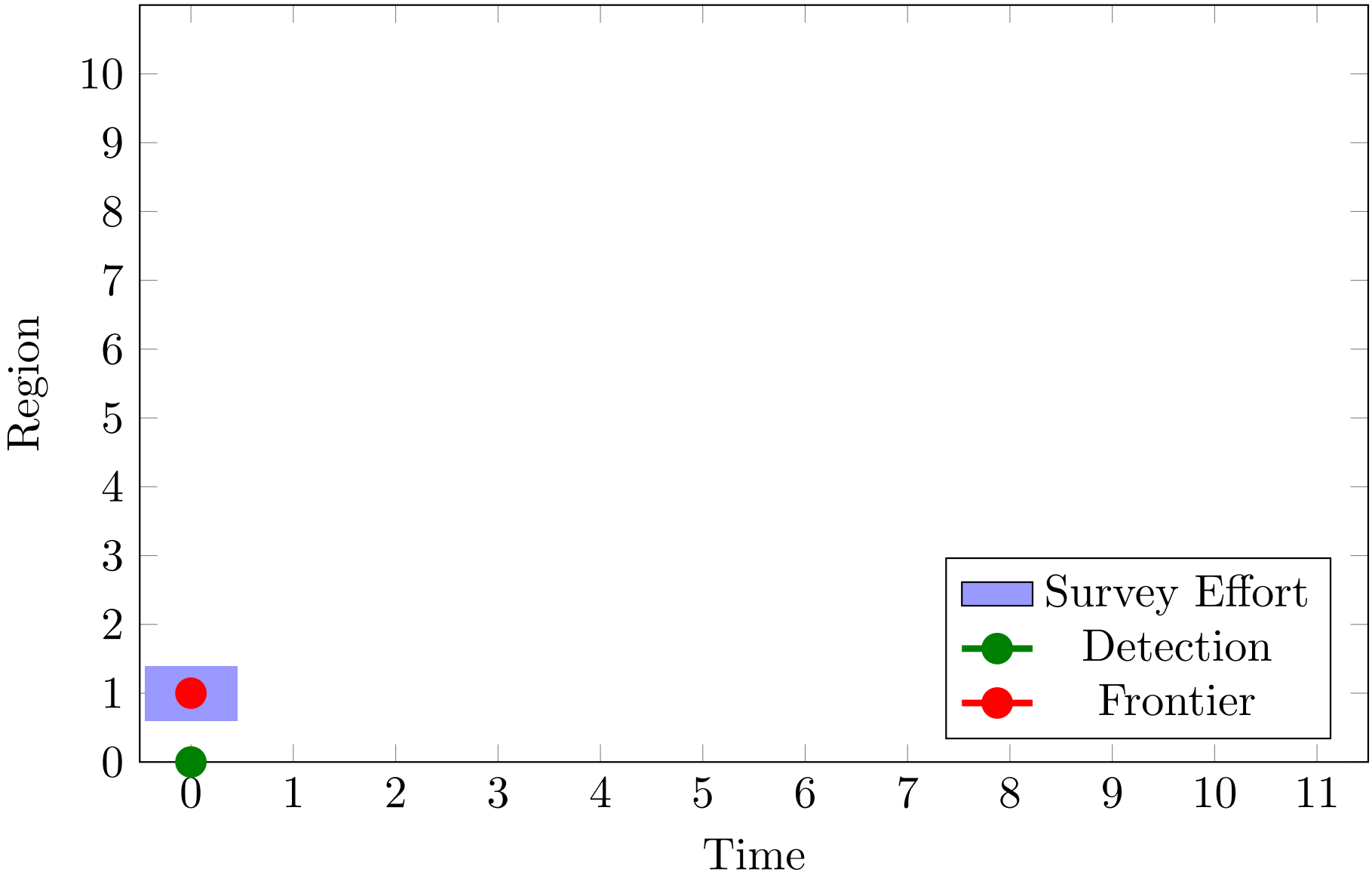
Case Study: Spotted Lanternfly



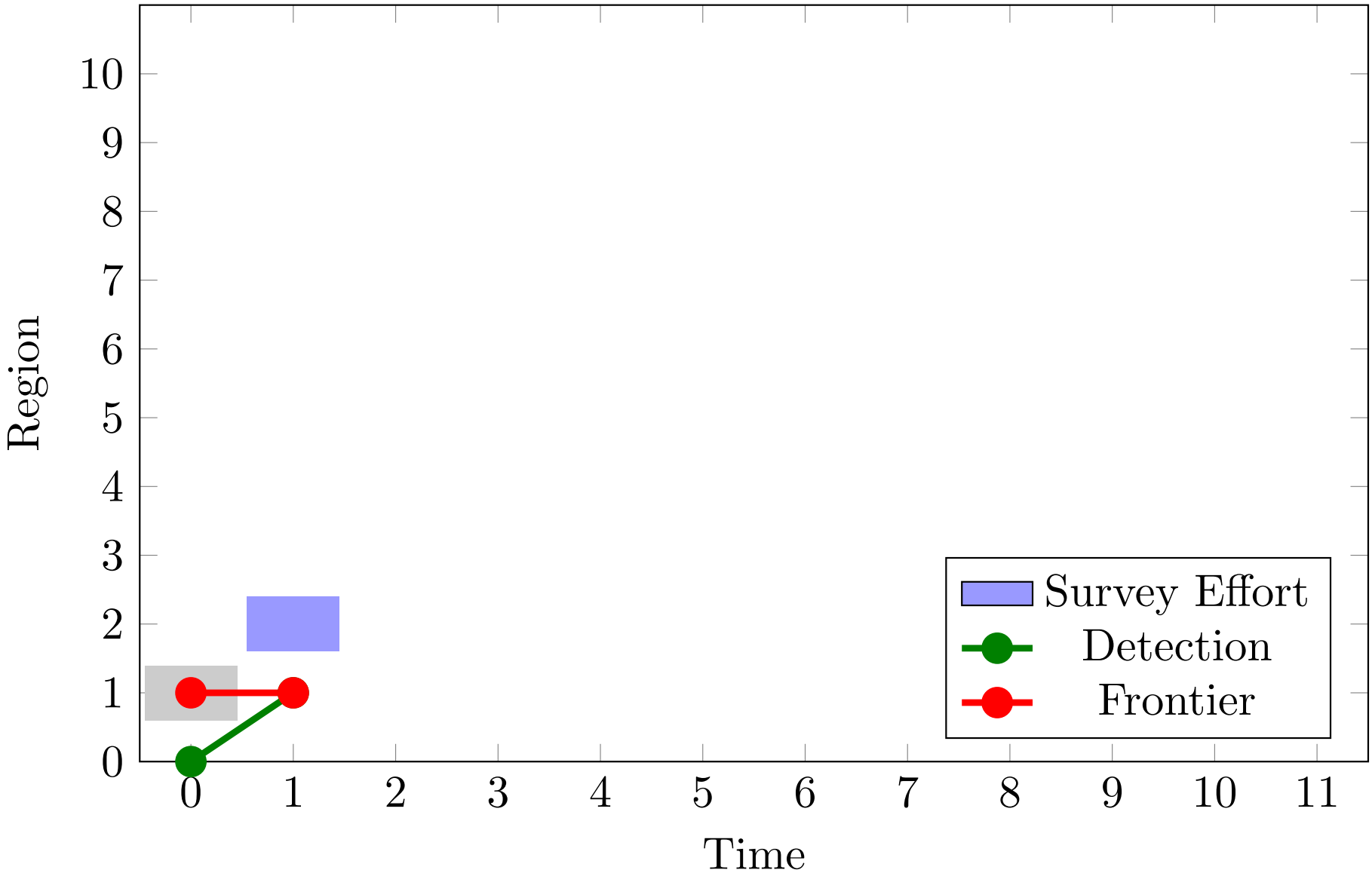
Figure: The spotted lanternfly

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- $\lambda B < 2\theta$: **Chase the frontier** by testing nearby the belief mean

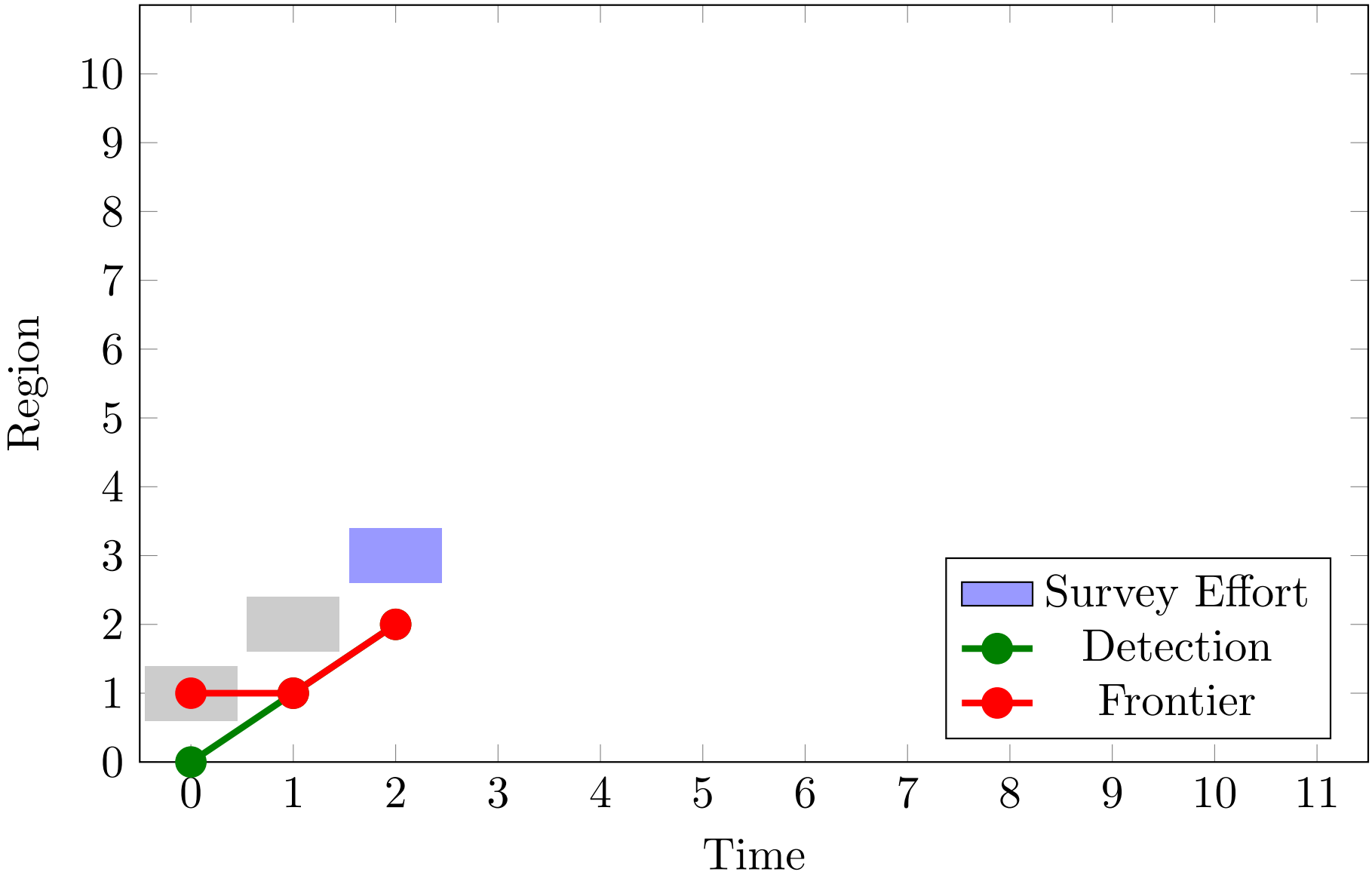
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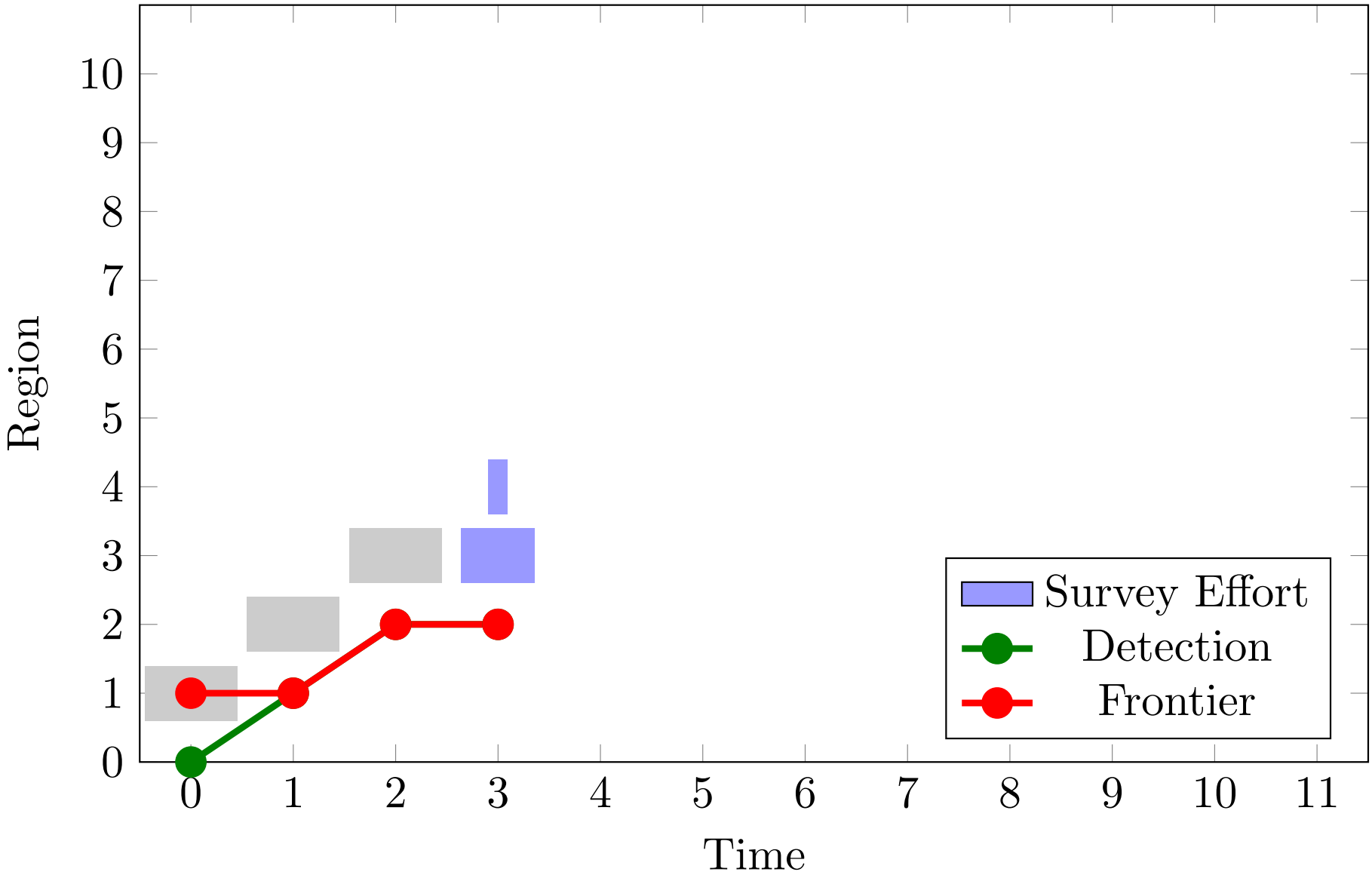
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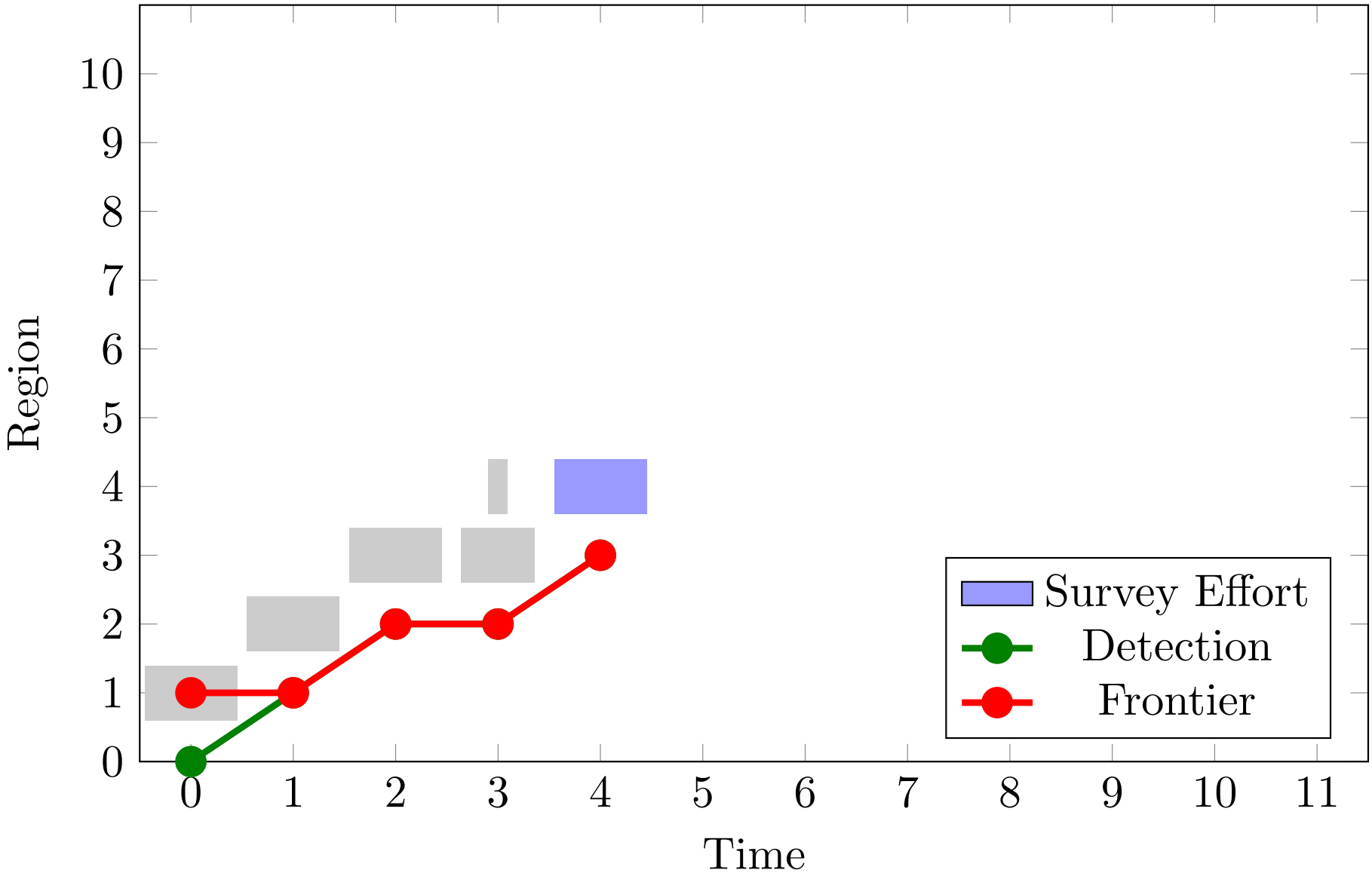
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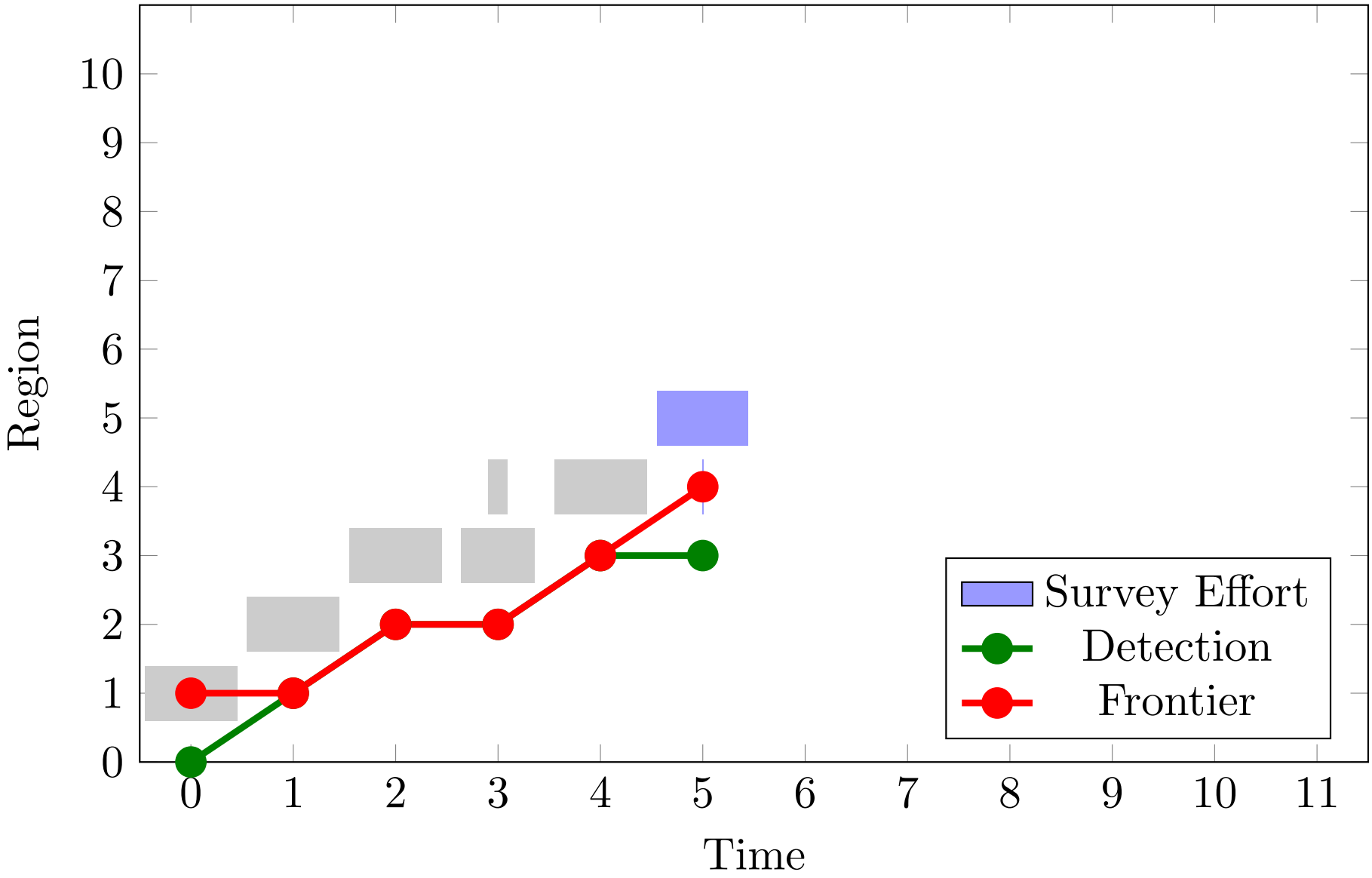
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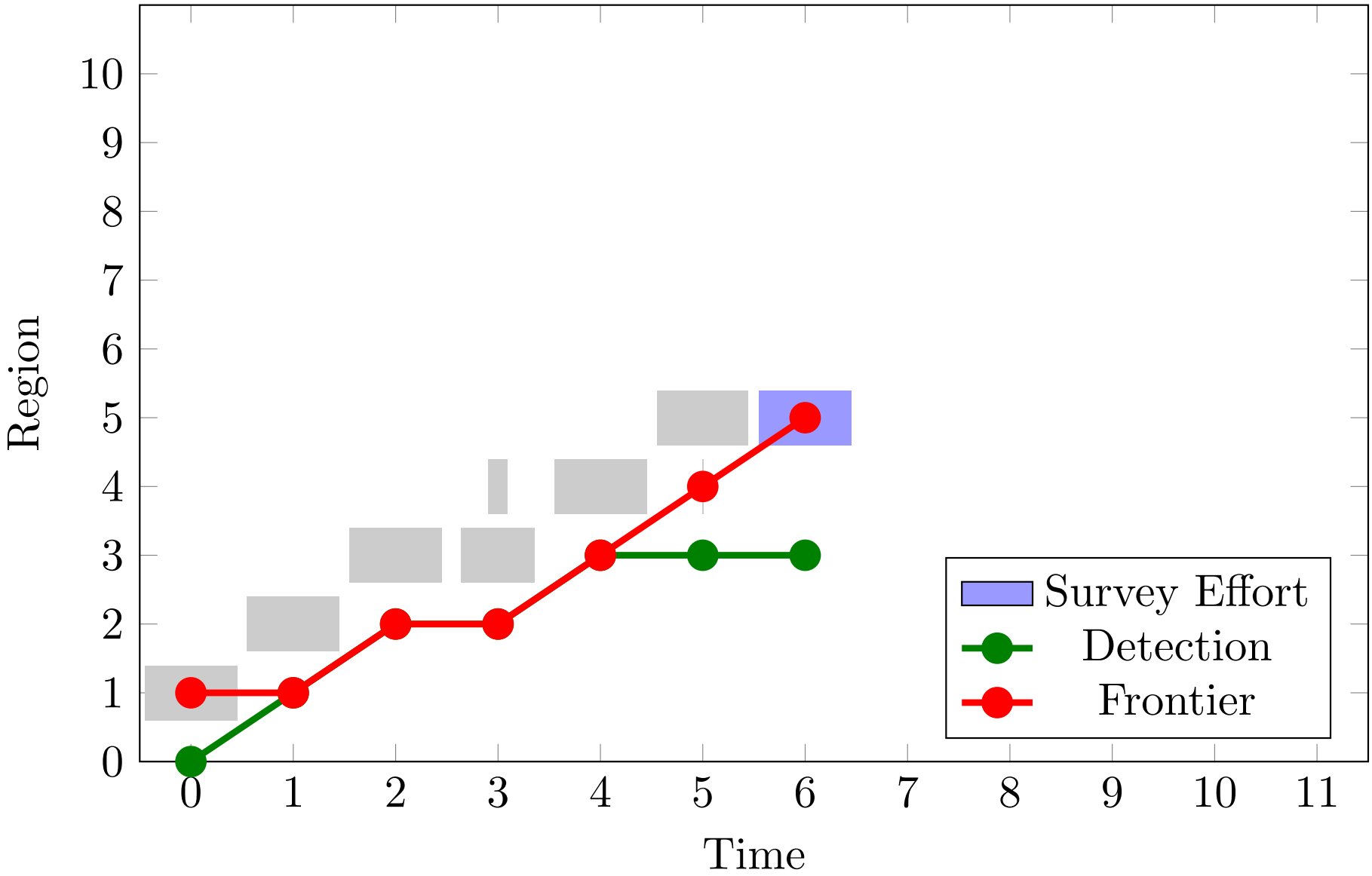
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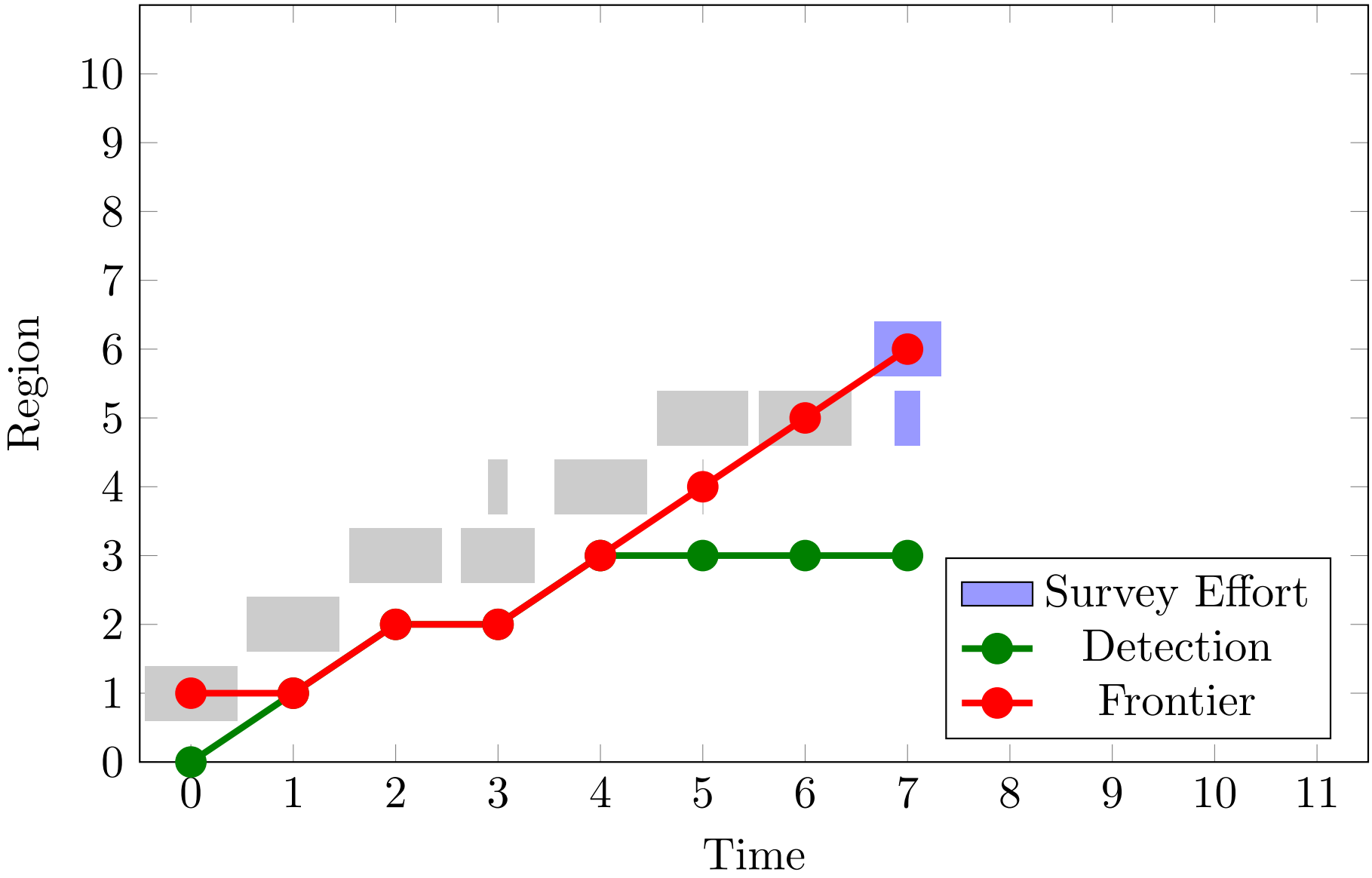
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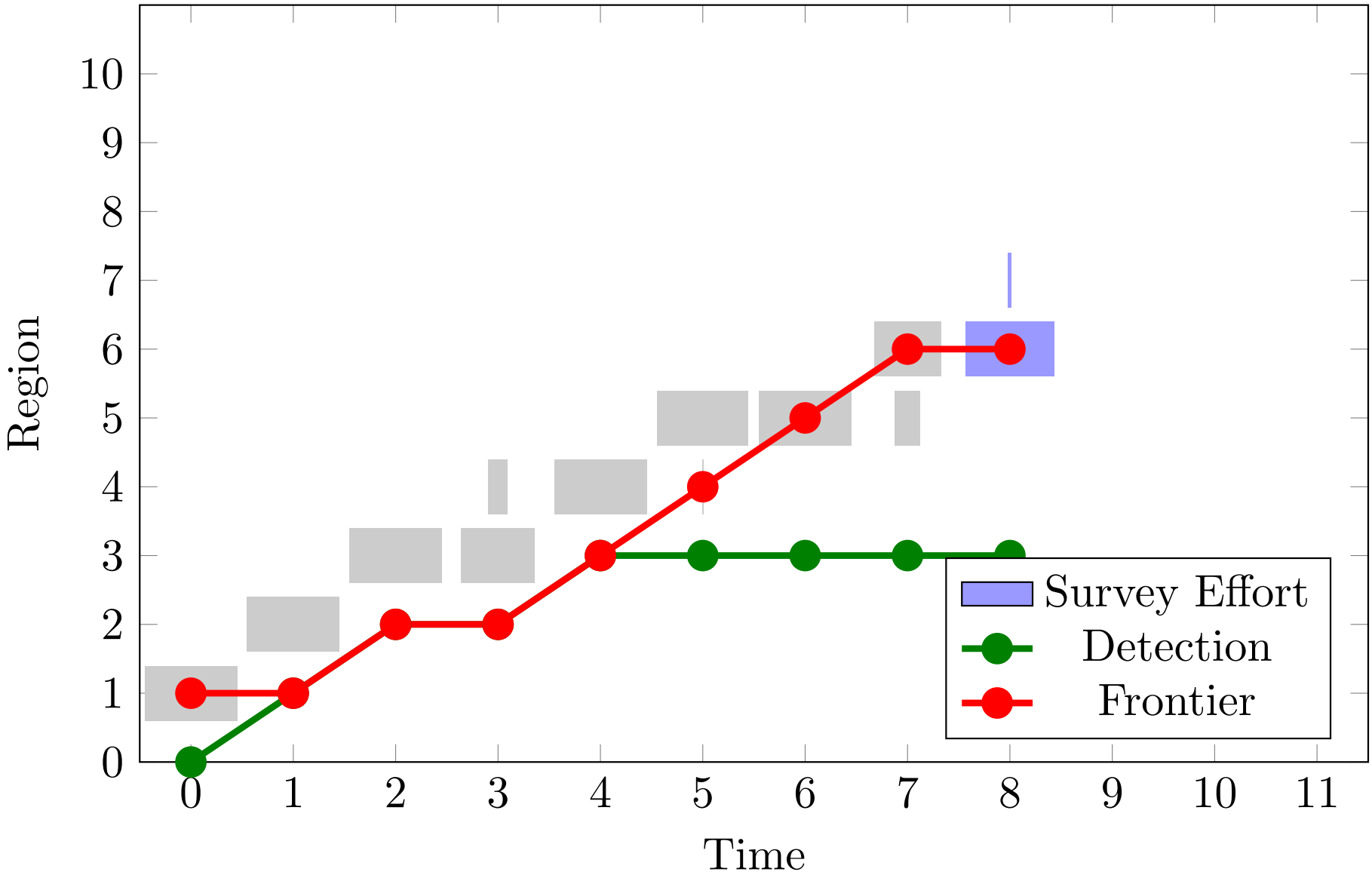
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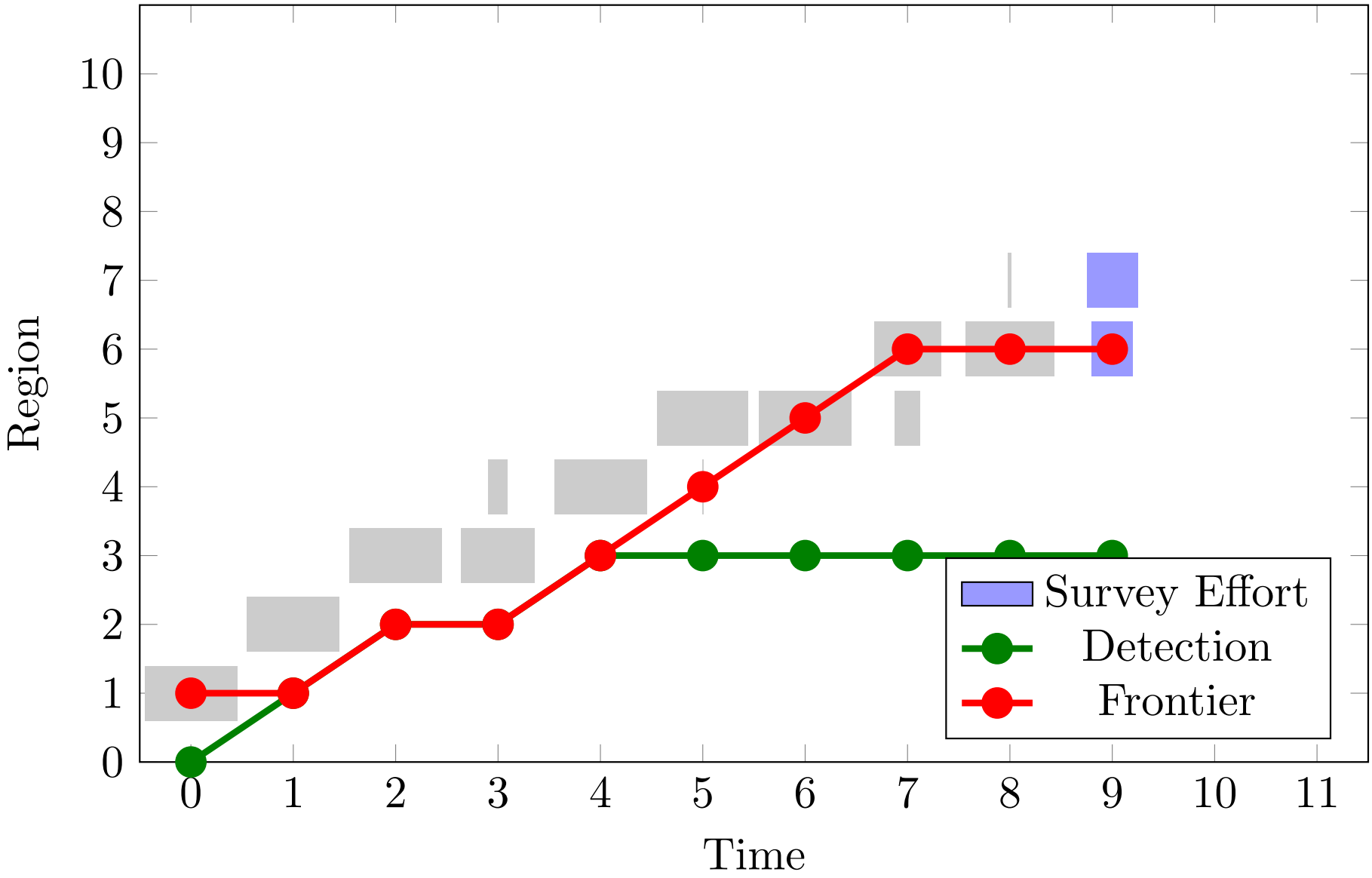
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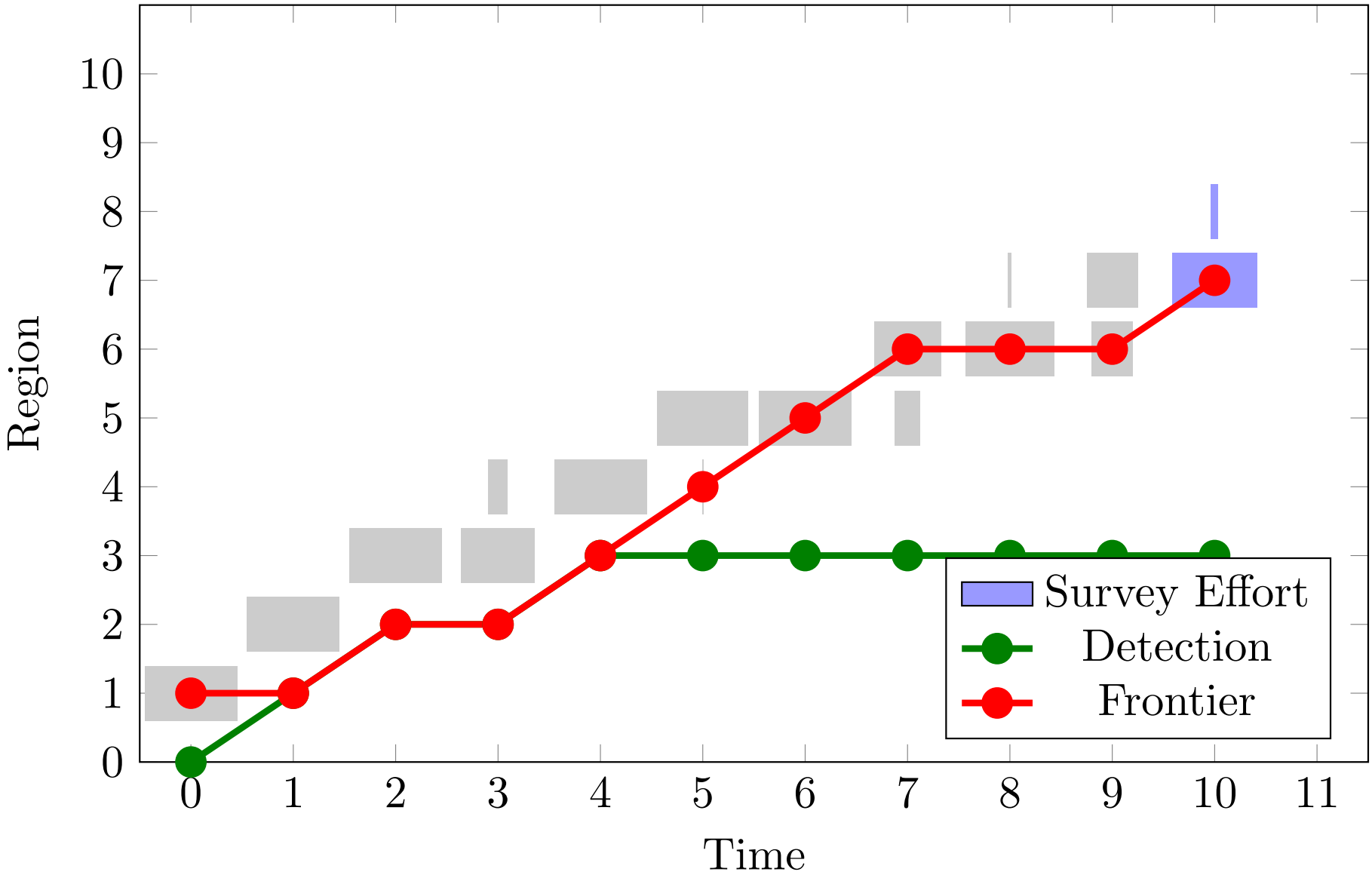
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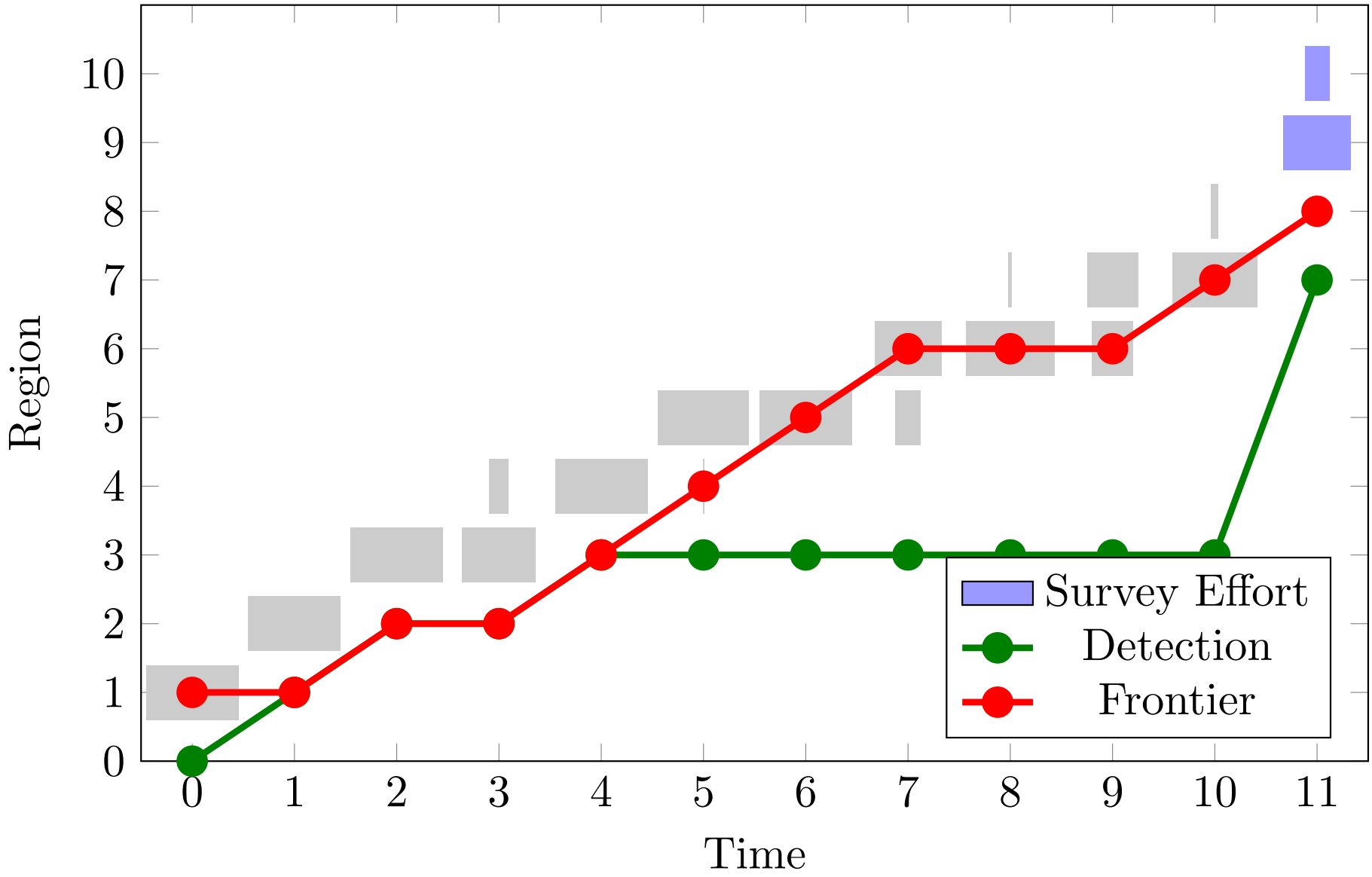
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Case Study: Hemlock Woolly Adelgid



Figure: The Hemlock Woolly Adelgid

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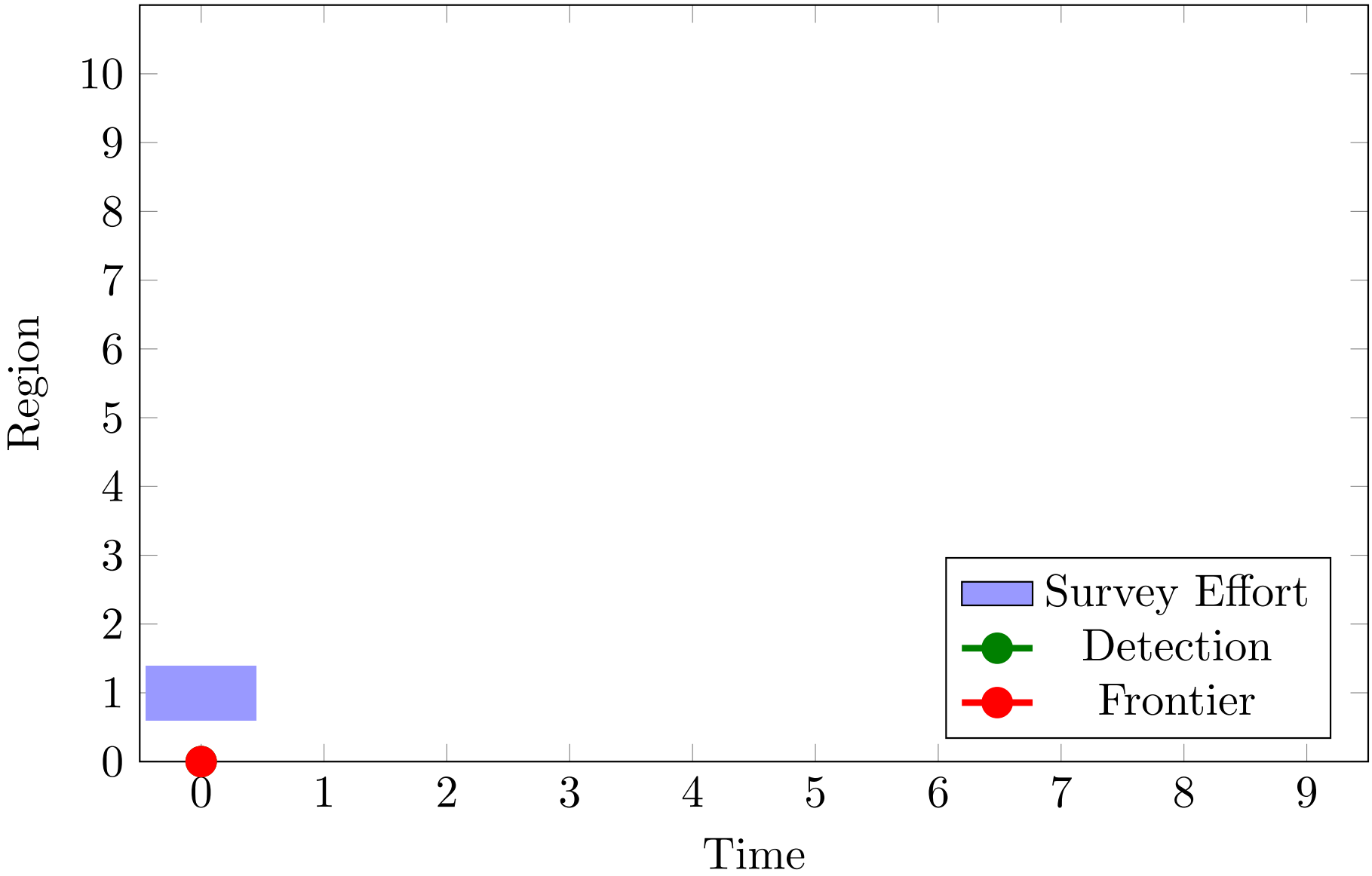
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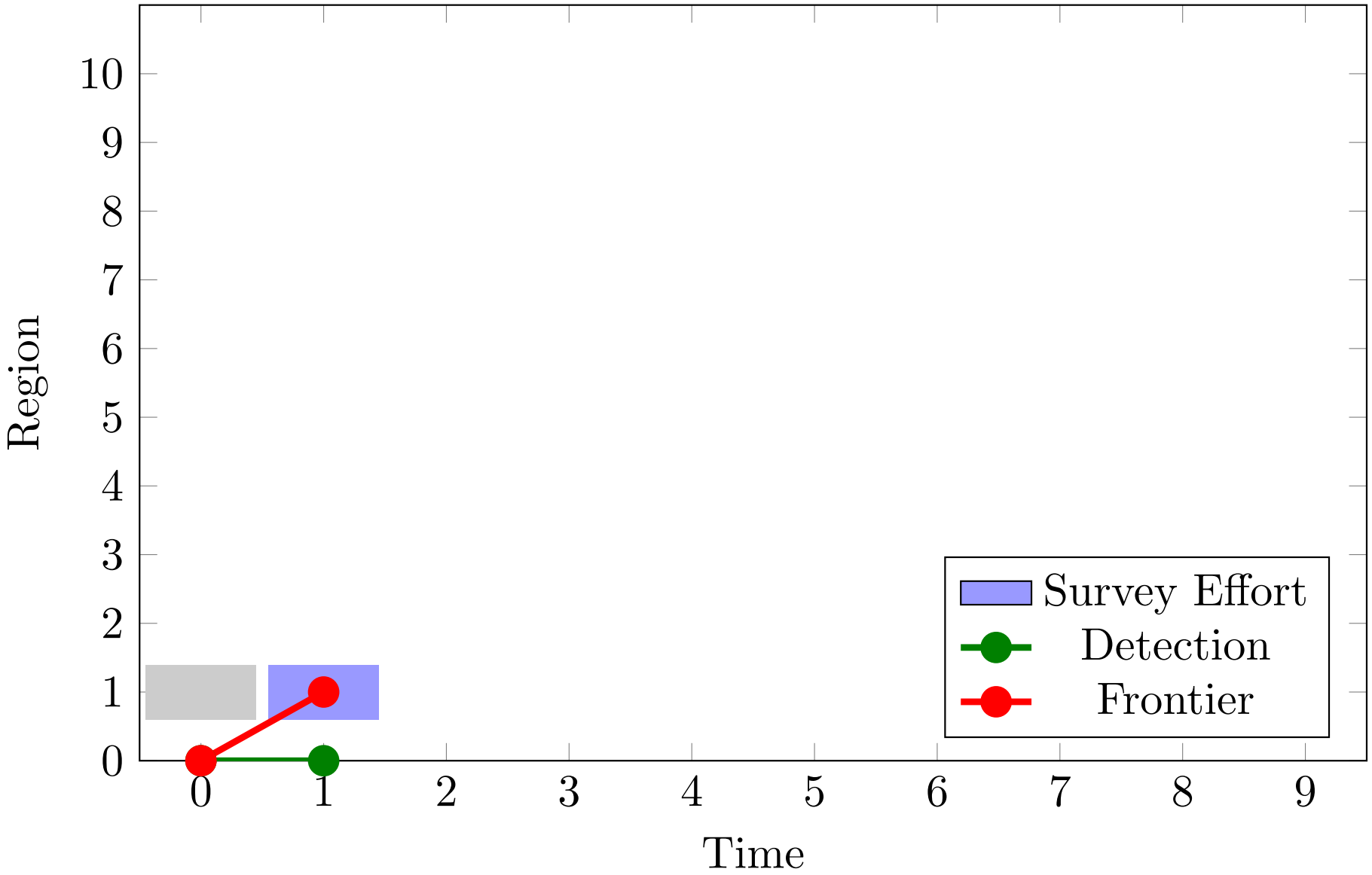
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- $\lambda B > 2\theta$: **Survey the next region**

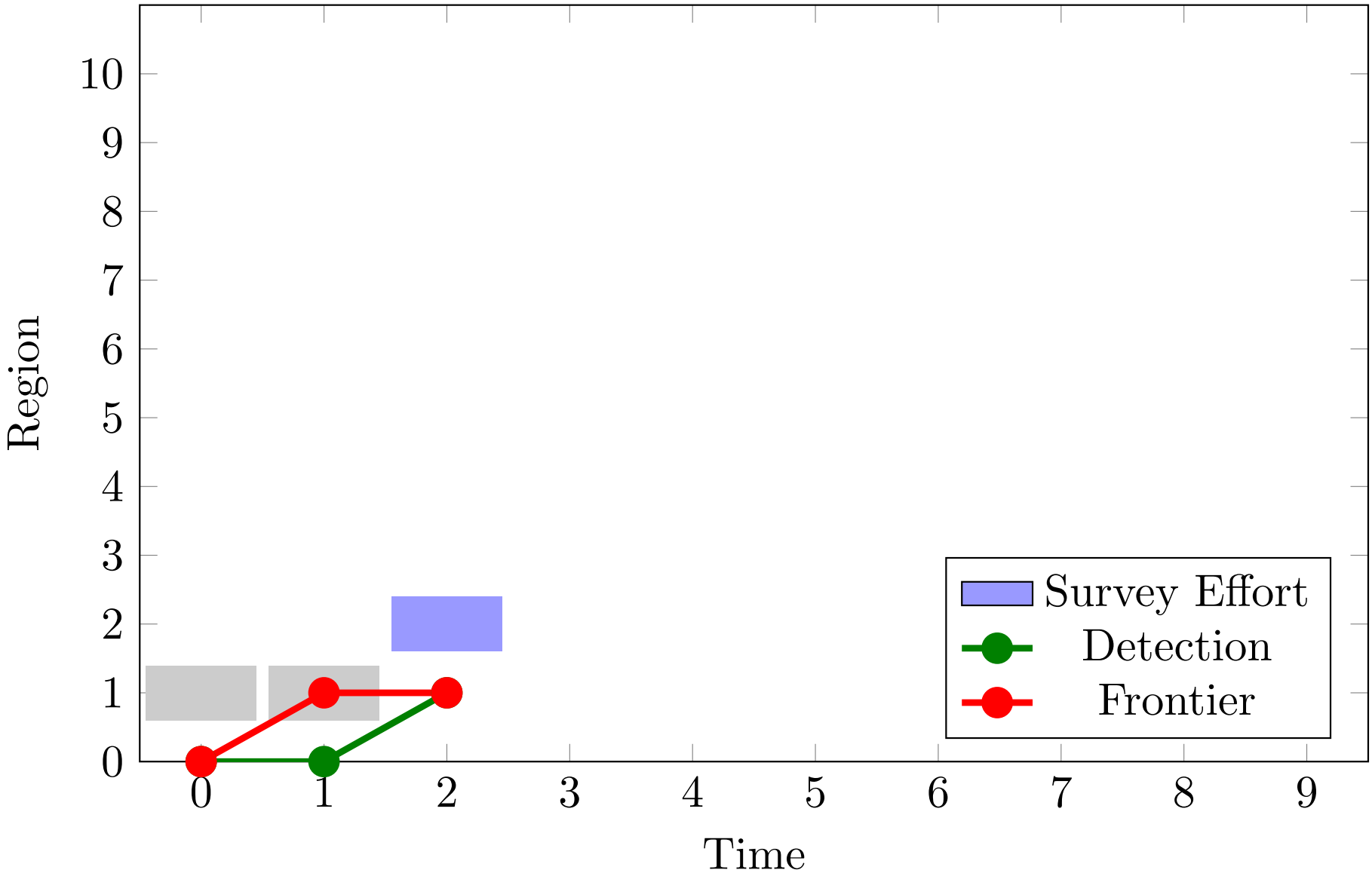
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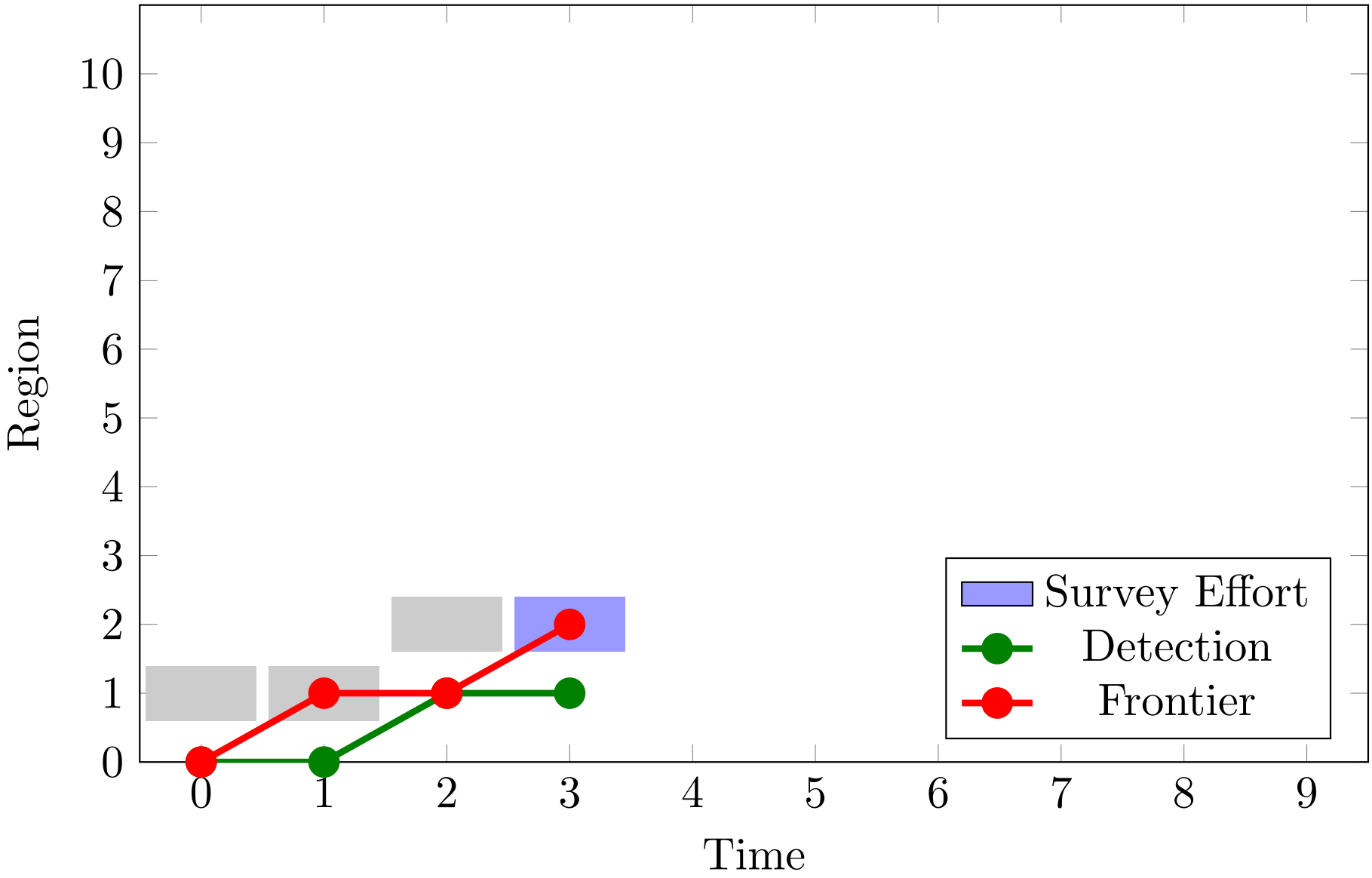
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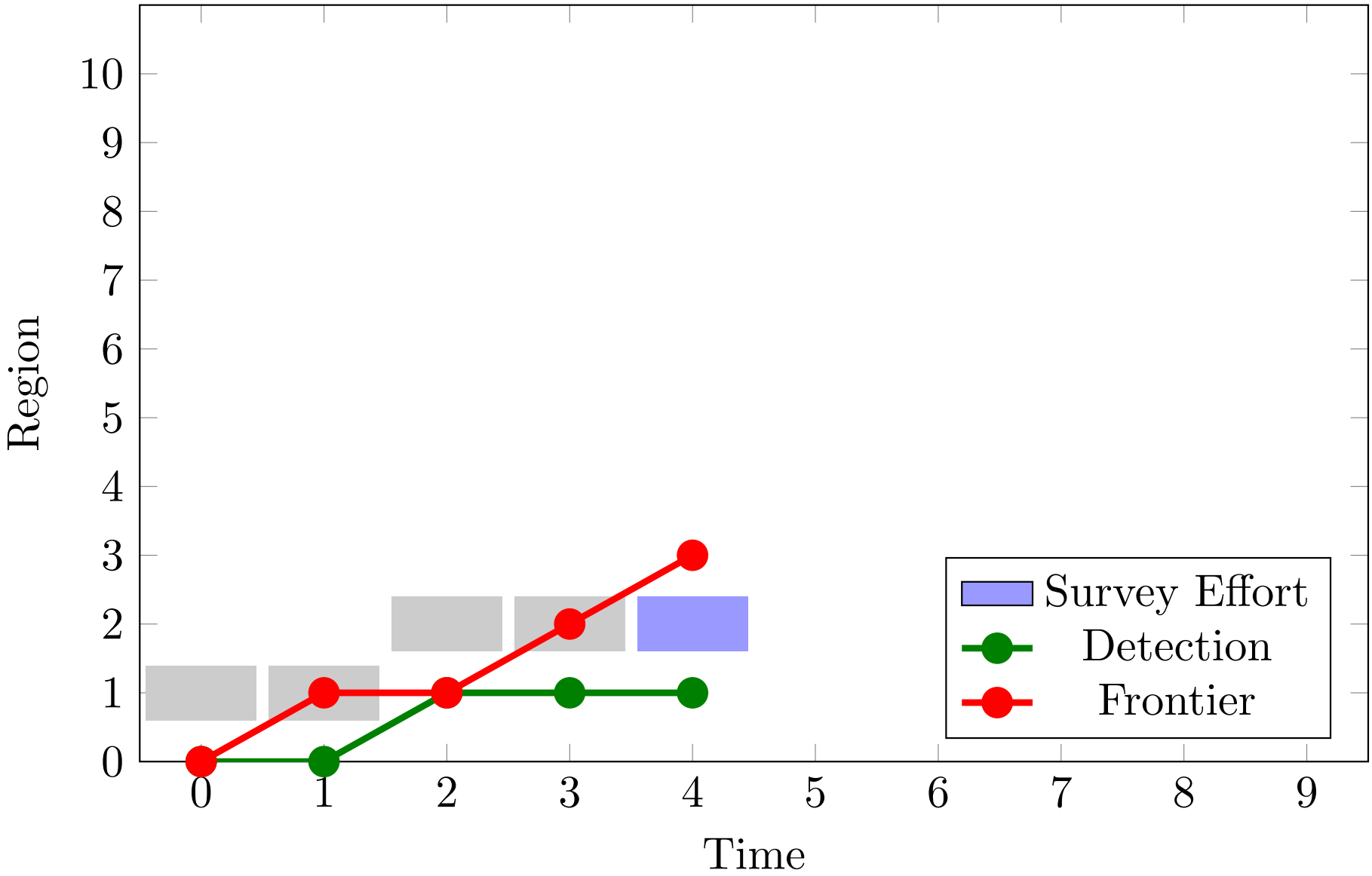
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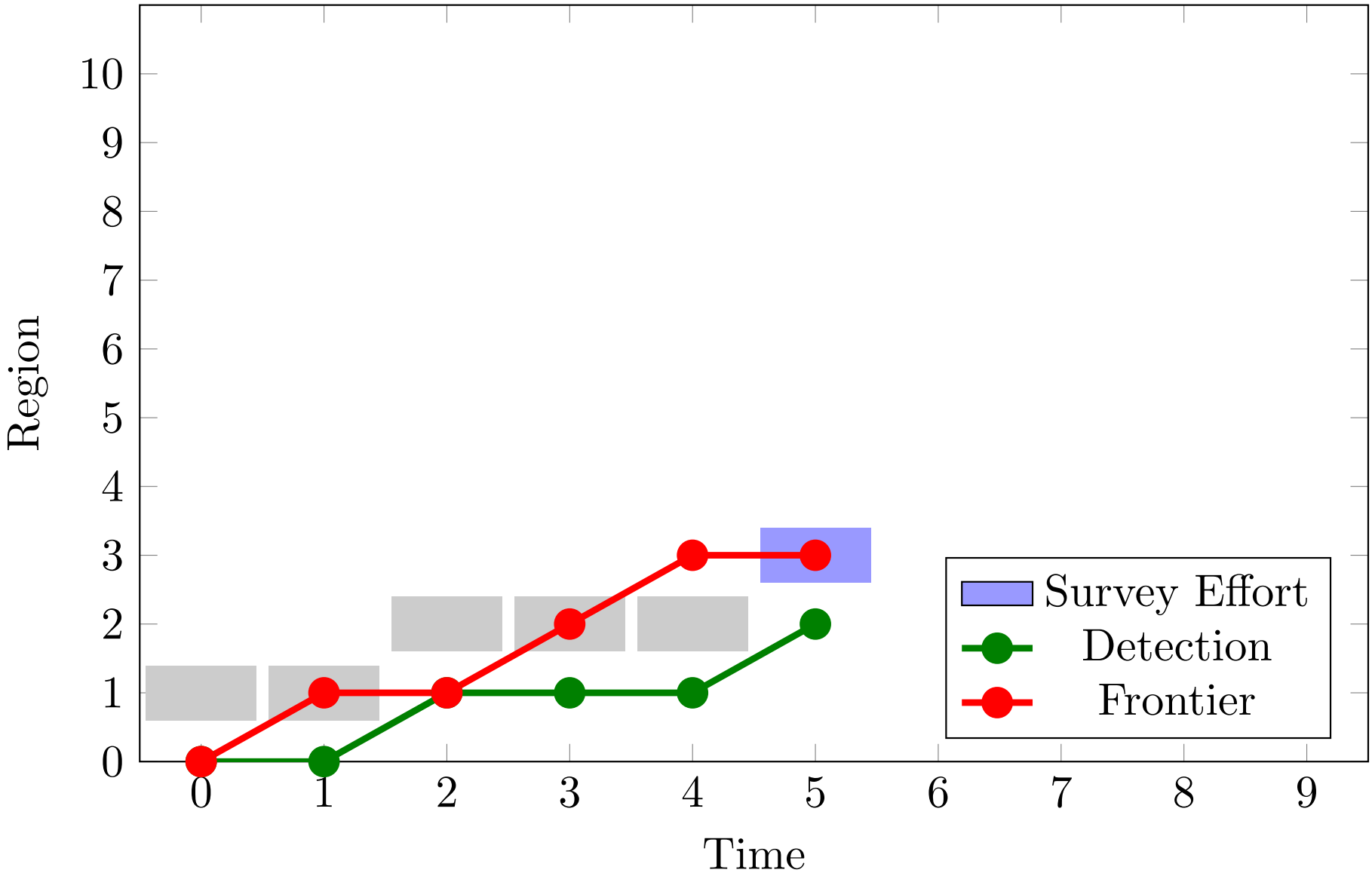
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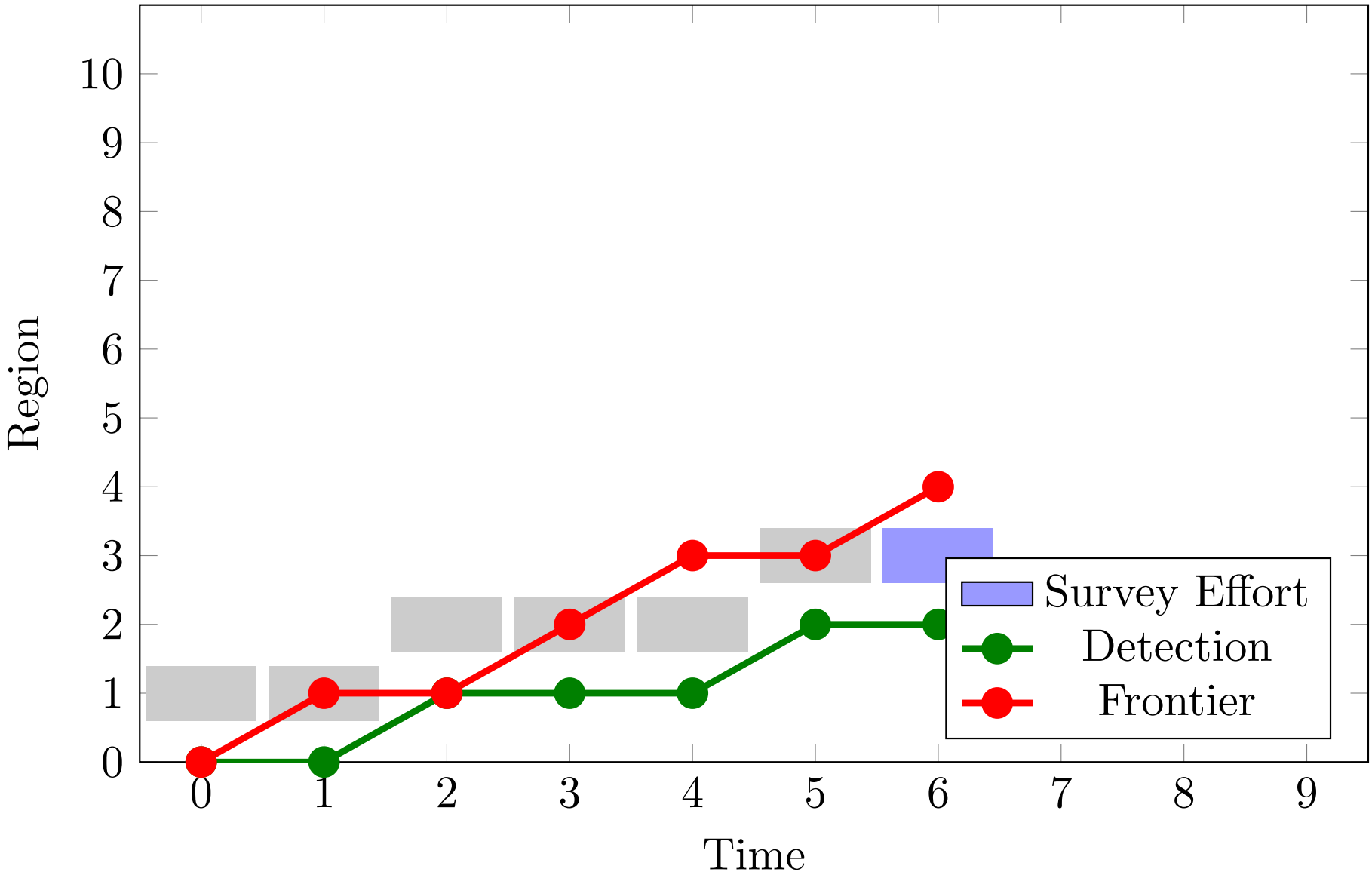
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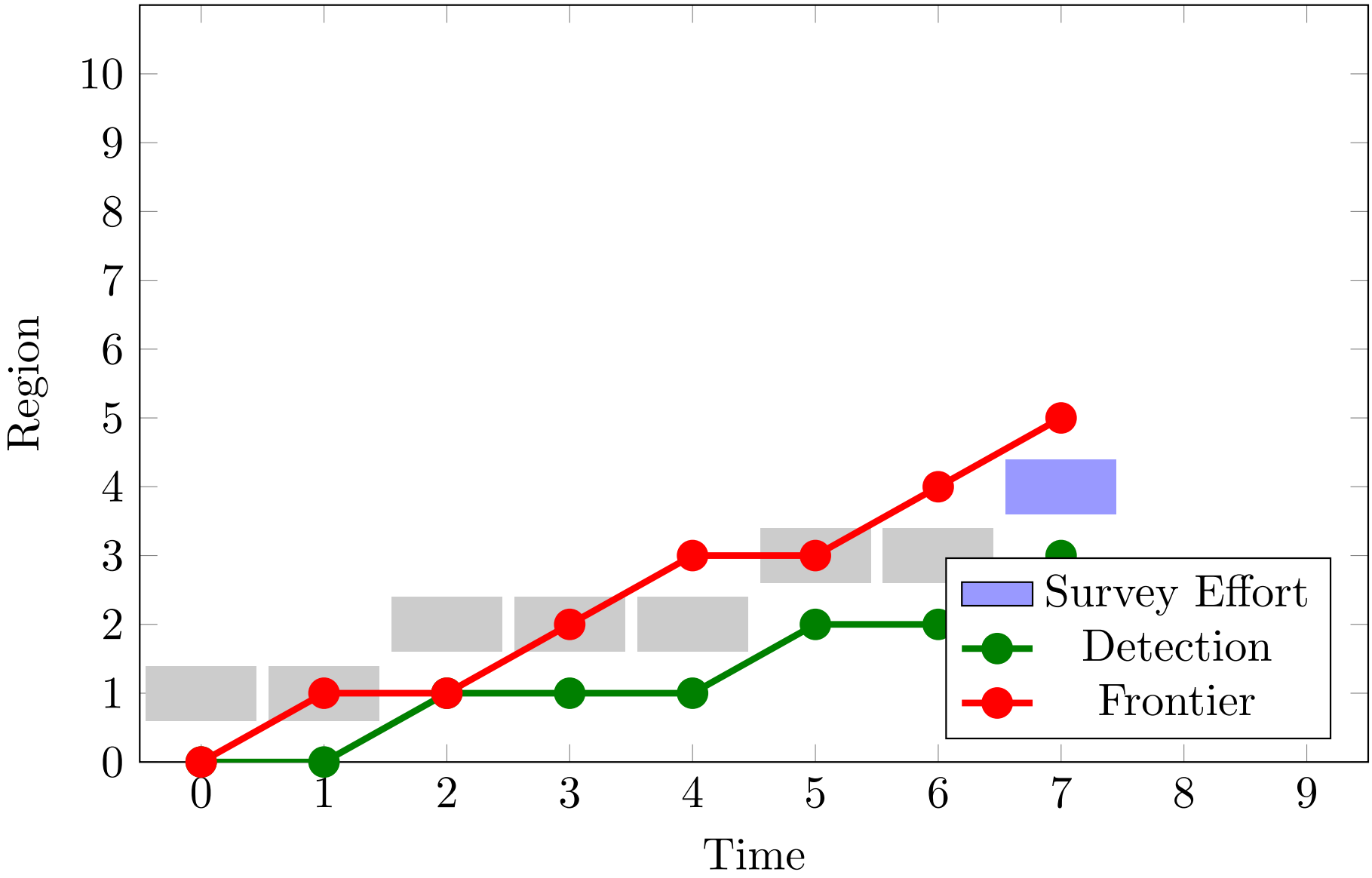
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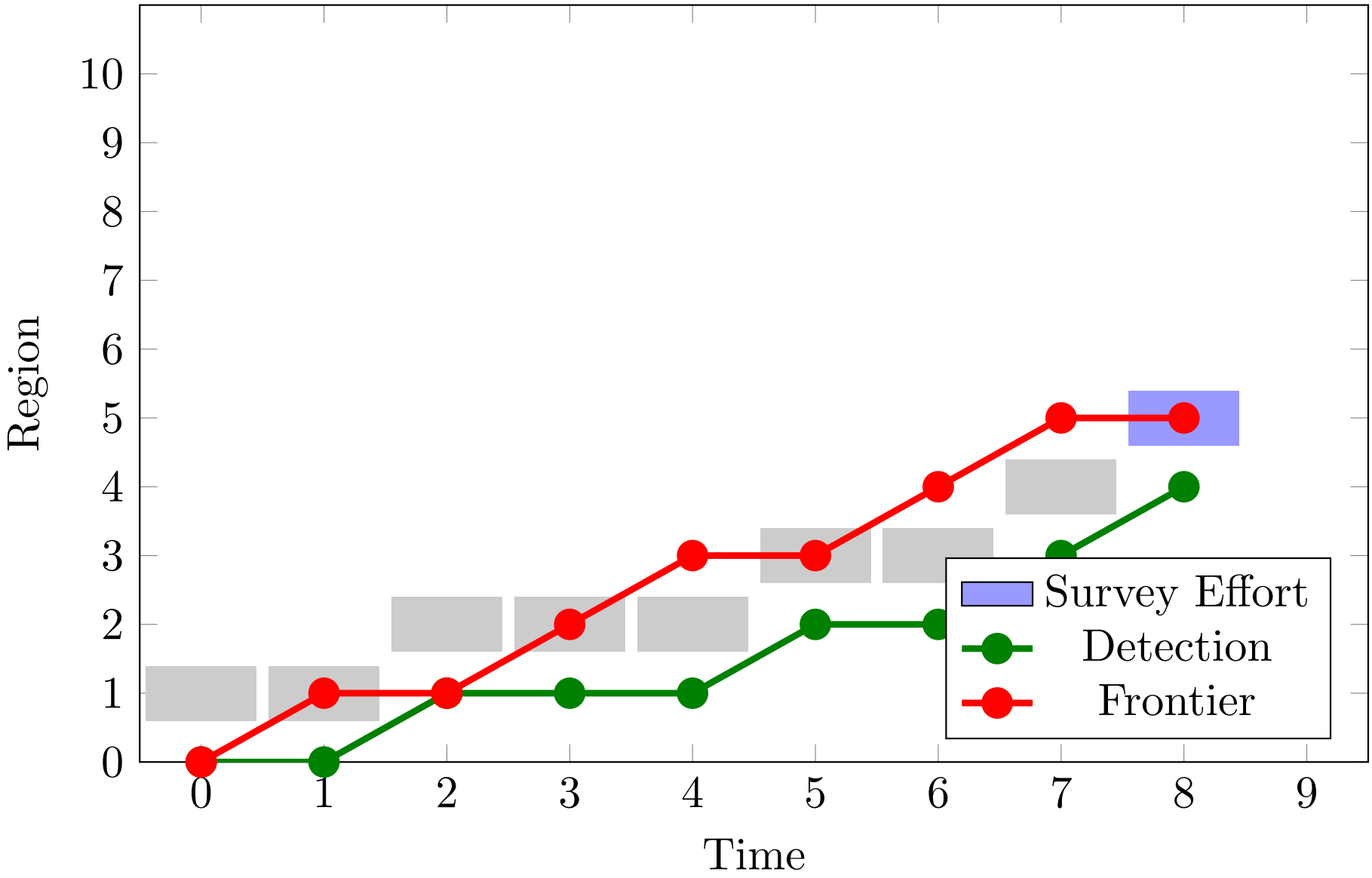
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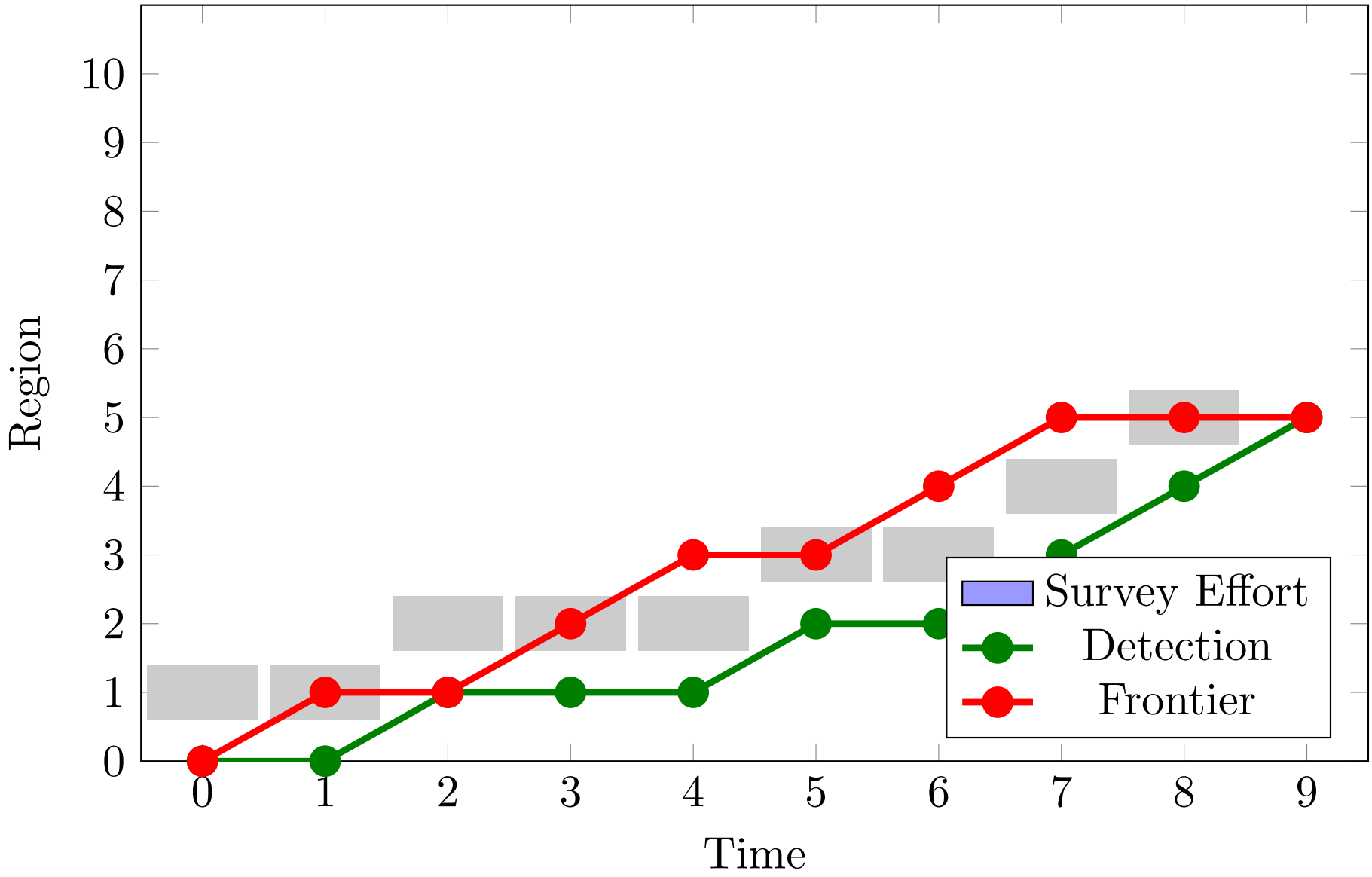
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